

VCDESIGN: CANDIDATE-CONDITIONED INVERSE MODELING FOR CELLULAR INTERVENTION DESIGN

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ABSTRACT

Virtual-cell modeling aims to predict cellular responses and prioritize interventions for experimental validation. Recent inverse models identify interventions for desired cellular transitions, but experimental planning requires deciding which candidates to test under a limited budget. We introduce VCDesign, which formulates intervention design as finite-budget ranking over a variable candidate set. Given starting and target states, VCDesign ranks interventions and evaluates selected candidates by held-out outcomes rather than recovery of a predefined intervention identity. This formulation accommodates retrieval, forward simulation, optimization, and direct scoring. For candidates with unmeasured responses, VCDesign-CED uses Candidate Effect Distillation to map biological knowledge to predicted perturbational effects, supervised by historical perturbation data. Across four CRISPRi Perturb-seq contexts, CED improves selection across held-out intervention identities, held-out contexts, and controlled response masking. Direct profile retrieval is more appropriate when candidate responses are available. Controlled analyses indicate that, within the tested regime, transferable design is limited primarily by candidate-side biological information and perturbation measurements rather than the additional model capacity explored. These findings clarify how response availability shapes solver choice and distinguish context transfer from response-unseen intervention transfer. VCDesign advances virtual-cell modeling from prediction and intervention identification toward cellular intervention design evaluated by the experimental decisions it supports.

1 INTRODUCTION

Virtual-cell models have advanced rapidly as tools for modeling cellular perturbations, but most work still centers on a forward problem: given an intervention, predict how the cell will respond (Lotfollahi et al., 2019; 2023; Bunne et al., 2023; Roohani et al., 2024; Cui et al., 2024; Yu et al., 2025; Adduri et al., 2026). Such prediction is essential for understanding perturbational effects, yet it is not the same as the decision faced in experimental planning. In practice, researchers often start from a cellular state they wish to change, have a desired state they hope to reach, and must choose among many possible interventions under a limited experimental budget. The central question is therefore which interventions are most worth testing first. This problem is distinct from forward prediction, and it is also not fully captured by identifying a plausible driver from a predefined output space. A model may accurately predict the consequence of a specified intervention, or identify a plausible intervention, without determining which experiments should actually be prioritized. We therefore argue that cellular intervention modeling should progress from prediction and identification to design, with the model ultimately evaluated by the experimental decisions it supports.

This distinction is particularly clear in recent inverse-modeling work. Methods such as CellNavi (Wang et al., 2025) move beyond forward prediction by using a starting and a desired cellular state to identify genes that may drive the transition. This is an important step from predicting intervention outcomes toward searching for interventions, but identifying a driver is not the same as designing an experiment. Experimental design rarely asks for recovery of one predefined correct identity. Instead, researchers face a candidate library whose composition may depend on experimental constraints, several interventions may be useful, and only a limited subset can be tested. Response retrieval, forward enumeration, and intervention optimization likewise approach this broader prob-

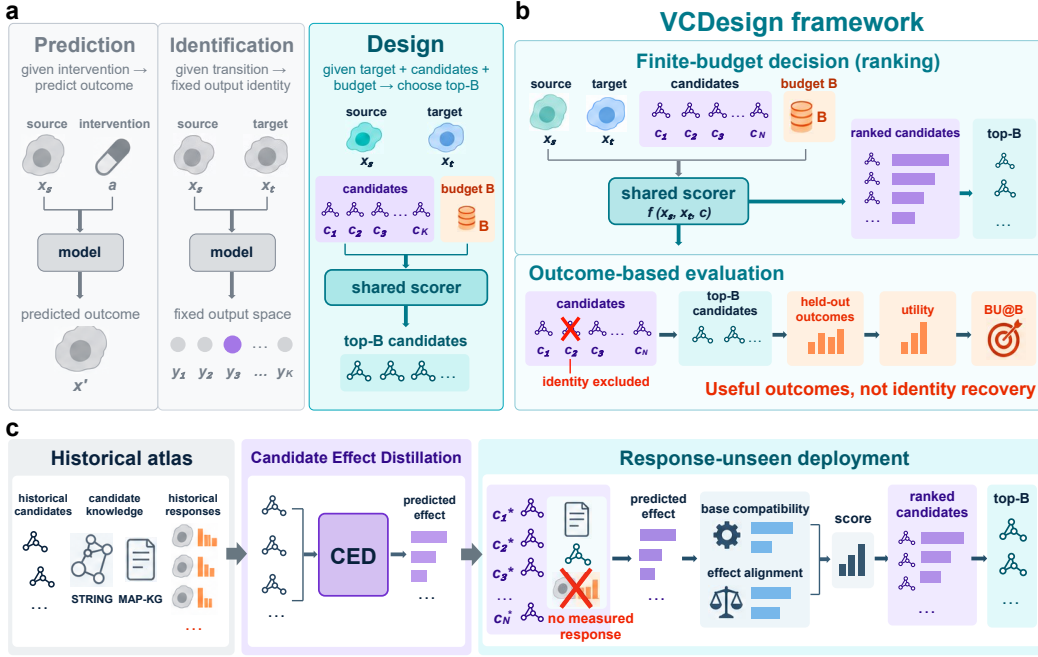


Figure 1: **The VCDesign framework and its VCDesign-CED reference realization.** **a**, Prediction maps an intervention to an outcome, and identification maps a transition to a fixed output space; design ranks a variable candidate set toward a target under a budget. **b**, One shared scorer $f(x_s, x_t, c)$ ranks every candidate, and the top- B prefix is evaluated by outcomes measured in a held-out replicate view, with the query’s own identity excluded. **c**, CED learns an effect predictor from historical responses and STRING and MAP-KG features; a candidate without a measured response is scored by base compatibility plus the alignment of its predicted effect with the requested transition.

lem through different computational routes (Gonzalez et al., 2025; Nourreddine et al., 2024; Han et al., 2025; Zeng et al., 2025; Cao et al., 2025; Peng et al., 2026; Liang et al., 2026). We therefore view them as different solvers for a higher-level design problem: what matters is not whether the answer is produced by classification, retrieval, simulation, or optimization, but how the method prioritizes experiments from the candidate set that is actually available.

Based on this view, we introduce VCDesign (Figure 1), which defines cellular intervention design directly in terms of the downstream experimental decision. Given a starting state, a desired target state, a variable set of feasible candidate interventions, and an experimental budget, the task is to rank the candidates and select a finite prefix for experimental validation. Candidate vocabulary size and composition are therefore part of the decision itself rather than dimensions fixed by a model output head. The evaluation must change accordingly. A desired cellular transition can often be achieved by more than one useful intervention, so design quality should not be reduced to recovering the intervention that generated the query. VCDesign instead removes that intervention from the candidate pool and evaluates the remaining candidates by independently measured held-out outcomes, directly assessing the quality of the experiments prioritized under the available budget. This decision-and-evaluation interface allows retrieval, forward simulation, optimization, and direct scoring to be studied within the same formulation, while also separating generalization to a new cellular context from generalization to an intervention whose own perturbational response has not been used for effect learning.

Within this formulation, a key question is not only how a solver is parameterized, but what information is available when the decision must be made. If a candidate intervention has already been experimentally profiled, its measured response provides direct evidence for ranking it. In a more prospective setting, however, an intervention must be prioritized before its own response is available. We refer to this setting as response-unseen intervention design and introduce VCDesign-CED as a reference realization. Candidate Effect Distillation (CED) uses historical perturbation responses as supervision to learn how deployment-available candidate knowledge relates to pertur-

bational effects, enabling previously unmeasured candidates to be ranked without using their own measured responses at deployment. We deliberately keep this effect-transfer mechanism simple in order to study a more fundamental question: when candidate responses are unavailable, is transferable design primarily limited by model capacity, by the biological information carried by candidate representations, or by the perturbation measurements available for learning?

Our contributions are threefold:

- **A decision-centric formulation of cellular intervention design.** We formulate intervention design as finite-budget ranking over a variable candidate set, allowing retrieval, forward simulation, optimization, and direct scoring to be understood and compared through a common downstream decision.
- **Outcome-based evaluation and a response-unseen realization.** We evaluate selected interventions using independently measured candidate outcomes rather than recovery of a single intervention identity, distinguish context transfer from response-unseen intervention transfer, and introduce VCDesign-CED as a concrete realization of the latter.
- **A controlled diagnosis of transferable design.** We systematically study how candidate-response availability, candidate information, perturbation measurements, and model capacity affect downstream design, clarifying the current limitations of response-unseen transfer.

2 RELATED WORK

Forward perturbation prediction. Virtual-cell models have primarily focused on predicting molecular responses to specified interventions. scGen and CPA model perturbation-induced state changes in learned latent spaces (Lotfollahi et al., 2019; 2023). CellOT learns transport maps between cellular populations, while GEARS incorporates gene relationships to predict unseen perturbation combinations (Bunne et al., 2023; Roohani et al., 2024). More recent methods extend perturbation prediction through pretrained cellular representations, as in scGPT and CellFM (Cui et al., 2024; Zeng et al., 2025), or transfer to unseen perturbations and contexts, as in PerturbNet and State (Yu et al., 2025; Adduri et al., 2026). In their perturbation-prediction tasks, these methods share a forward formulation: given an intervention and cellular context, predict the resulting molecular state. Recent benchmarks show that common prediction metrics can be dominated by shared variation and may not isolate perturbation-specific information (Ahlmann-Eltze et al., 2025; Viñas Torné et al., 2026).

A forward model can also support intervention design by simulating candidates and ranking their predicted outcomes against a target. The prediction task alone, however, does not specify the candidate set, experimental budget, or utility used to evaluate the selected interventions. VCDesign makes these decision components explicit and accommodates forward simulation within a shared ranking and outcome-based evaluation interface.

Inverse perturbation and design. A complementary line of work starts from a desired cellular state or transition and searches for interventions that could induce it. Connectivity Map and LINCS support perturbation retrieval by matching or reversing gene-expression signatures against experimentally measured profiles (Lamb et al., 2006; Subramanian et al., 2017). CellNavi predicts genes that drive transitions between cell states, while PDGrapher predicts therapeutic perturbations toward a desired healthy state (Wang et al., 2025; Gonzalez et al., 2025). CHEF learns to predict perturbation recipes for desired transcriptional states (Nourreddine et al., 2024). PAIRING matches perturbation effects to a desired transition in latent space, whereas RevPert ranks candidate drivers using a predicted response gallery (Han et al., 2025; Liang et al., 2026). CASCADE and U-Pert approach intervention design through optimization in learned regulatory models or perturbation dynamics (Cao et al., 2025; Peng et al., 2026). These methods provide complementary routes from desired outcomes to candidate interventions.

These inverse approaches differ in their output spaces, access to candidate responses, and evaluation objectives. VCDesign defines a common downstream decision: rank a supplied candidate set under a finite budget and evaluate the selected interventions by held-out outcomes. This formulation separates the decision from the computational route used to solve it. Its response-unseen regime

further tests whether a candidate can be prioritized when its own perturbational responses have not supervised the effect predictor in any training context.

3 THE VCDESIGN FRAMEWORK

VCDesign connects the experiments to prioritize, the solver that ranks them, and the outcomes used to judge that choice (Figure 1).

3.1 DESIGN UNDER A FINITE BUDGET

A query $q = (x_s, x_t)$ specifies source and target states, with transition $\mathbf{d}_q = x_t - x_s$. Given a feasible candidate set $\mathcal{C}_q$ and budget B , a solver ranks candidates and selects the first B for testing (Mehrjou et al., 2022). Retrieval can match measured responses; a forward model can simulate outcomes; an optimizer can search over interventions; and a direct scorer can rank candidate features. Each route supplies an ordering of the feasible set for the same downstream decision. The formulation therefore separates what makes an intervention useful from how a solver identifies it.

Our reference realization uses a shared score $s(q, c) = f(x_s, x_t, c)$ and candidate knowledge $\mathbf{k}_c$ available before experimental profiling. This lets the candidate library vary without changing a fixed model output vocabulary.

3.2 EVALUATION BY MEASURED OUTCOMES

In our CRISPRi benchmark, useful interventions reproduce the direction of a desired transcriptomic change. Each screen is split into batch-disjoint views $v \in \{1, 2\}$; a query formed in one view is evaluated in the other. The response $\mathbf{r}_g^{(v)}$ is the log-normalized pseudobulk of knockdown g minus its view-specific control. For query identity g_q , utility is

$$U(q, c) = \cos(\mathbf{r}_{g_q}^{(\bar{v}_q)}, \mathbf{r}_c^{(\bar{v}_q)}), \quad (1)$$

where $\bar{v}_q$ is the held-out evaluation view. We exclude g_q from the pool, rewarding other interventions that produce useful outcomes; alternative utilities are compared in Section E, and other applications can use their own measured outcomes.

Budgeted utility measures the value recovered within B experiments. Let $\text{RU@}B$ be their discounted mean utility, with rank weights $1/\log_2(j+1)$ (Järvelin & Kekäläinen, 2002). We normalize against random selection and the best ranking under measured utility:

$$\text{BU@}B(q) = \frac{\text{RU@}B(q) - \text{RU}_{\text{rand}}(q)}{\text{RU}^*\text{@}B(q) - \text{RU}_{\text{rand}}(q)}. \quad (2)$$

Zero denotes random selection in expectation; one denotes the best ordering of that pool under U . We report $\text{BU@}20$ and the budget average $\overline{\text{BU}} = \frac{1}{3}(\text{BU@}10 + \text{BU@}20 + \text{BU@}50)$. Mean selected utility $\text{MU@}B$ and high-value hit counts $\text{HvHit@}B$ complement this score; high-value candidates are in their pool’s top 5% by utility. Queries must have reproducible responses across views (definitions and eligibility test in Sections A and A.1).

3.3 CONTEXT AND RESPONSE GENERALIZATION

A new cellular context and an unmeasured candidate pose distinct transfer problems. *Context transfer* withholds the evaluation context from training; the candidate may still have measured responses in other contexts. *Response-unseen transfer* removes an evaluated candidate’s responses from the CED atlas in every training context. The base scorer remains fixed; its K562 training identities include about two thirds of the masked candidates, so masking tests the effect branch, not the whole system. Held-out identities, held-out contexts, and response masking thus distinguish what is new to the scorer from what is new to the effect predictor.

4 VCDESIGN-CED

When candidate responses are unavailable, the solver must infer which interventions could produce the desired transition. VCDesign-CED learns this relationship from historical perturbation screens. It combines an effect predictor learned from candidate knowledge with a direct candidate-conditioned scorer (Figure 1b,c).

4.1 LEARNING EFFECTS FROM CANDIDATE KNOWLEDGE

Candidate Effect Distillation (CED) uses measured responses as supervision for a mapping from biological knowledge to perturbational effects. Candidate features combine STRING physical-interaction information and MAP-KG features derived from protein representations (Szkarczyk et al., 2023; Feng et al., 2026; Lin et al., 2023).

CED represents training responses in a low-dimensional basis. Let the rows of $\mathbf{R}$ be training consensus responses and let $\mathbf{W}$ contain its 256 leading uncentered right singular vectors (Halko et al., 2011). Ridge regression maps standardized knowledge $\tilde{\mathbf{k}}_c$ to coefficients in this response basis:

$$(\hat{\mathbf{A}}, \hat{\mathbf{b}}) = \arg \min_{\mathbf{A}, \mathbf{b}} \sum_c \|\mathbf{W}^\top \mathbf{r}_c - \mathbf{b} - \mathbf{A} \tilde{\mathbf{k}}_c\|_2^2 + \lambda \|\mathbf{A}\|_F^2, \quad (3)$$

$$\hat{\mathbf{e}}_c = \mathbf{W}(\hat{\mathbf{A}} \tilde{\mathbf{k}}_c + \hat{\mathbf{b}}), \quad s_{\text{eff}}(q, c) = \cos(\mathbf{d}_q, \hat{\mathbf{e}}_c). \quad (4)$$

At deployment, $\hat{\mathbf{e}}_c$ is computed from knowledge alone and scored by alignment with the requested transition. This attribute-to-output mapping connects CED to zero-shot learning and learning with privileged information (Xian et al., 2019; Vapnik & Vashist, 2009; Lopez-Paz et al., 2016). Its linear form tests how much transferable information candidate features carry; richer maps are compared in Section 5.4.

Held-context results use unit-direction targets $\mathbf{r}_c / \|\mathbf{r}_c\|$ with the same ridge and fusion; K562 results use raw targets (Sections B and E.1).

4.2 DIRECT CANDIDATE SCORING AND FUSION

A dual encoder maps $(x_s, \mathbf{d}_q)$ and $\mathbf{k}_c$ to unit vectors $\mathbf{u}_q$ and $\mathbf{v}_c$, respectively (Radford et al., 2021; Karpukhin et al., 2020). Trained on K562 to prioritize high-utility candidates, it gives the compatibility score $s_{\text{base}}(q, c) = \tau \langle \mathbf{u}_q, \mathbf{v}_c \rangle$. We standardize each branch across the current candidate pool and add them with equal weight:

$$s(q, c) = z_q(s_{\text{base}}(q, c)) + z_q(s_{\text{eff}}(q, c)). \quad (5)$$

The direct branch learns which candidate features mark useful interventions; the effect branch predicts a response to compare with the requested transition (details in Section B).

In a new context, CED pools historical responses from the other contexts on their shared 6,550-gene axis; the evaluation context contributes no training response, and the K562-trained base scorer stays fixed.

5 EXPERIMENTS

The experiments ask how well VCDesign-CED selects interventions, when it is the right solver, and what limits it.

5.1 EXPERIMENTAL SETUP

We use CRISPRi Perturb-seq screens from K562 and RPE1 (Replogle et al., 2022) and HepG2 and Jurkat (Nadig et al., 2025). K562 supplies disjoint training, development and locked evaluation identities. RPE1, HepG2 and Jurkat are held out in turn as new contexts, with pools of 1,273, 764 and 1,450 candidates (Section A); K562, where the base scorer was trained, is analysed as a held context only in the appendix (raw-target masked gain +0.0005 $\overline{\text{BU}}$, Section D.11). In a held context, an evaluated candidate’s responses are either *masked* from the effect atlas in every training

Table 1: **Solvers, the information they need, and how well they select.** Median BU over query-view rows: $\overline{BU}$ on the locked K562 split, whose candidates neither branch has seen, and $BU@20$ in held-out contexts, with each candidate’s own response masked from the effect atlas or measured elsewhere. The K562-trained base scorer saw about two thirds of the masked candidates, so paired gains (Figure 2) isolate the effect branch. Oracle: 0.814 and 0.951. —: not applicable.

Solver	Interface	Held-out	Held-out contexts	
		identities	masked	measured
<i>Can score without a candidate-specific measured response</i>				
Random	none	−0.005	0.002	−0.005
Goal-blind prior	effect size, ignores target	0.181	0.170	0.194
CED effect only	forward simulation + match	0.125	0.294	0.396
Text-augmented ranker	direct scoring + text	0.385		
Base scorer	direct scoring	0.355	0.440	0.440
VCDesign-CED	direct scoring + effect	0.391	0.499	0.508
<i>Requires a candidate-specific measured response</i>				
CellNavi, official*	fixed-output identification		−	−0.020
CellNavi, native*	fixed-output identification		−	0.015
Profile retrieval	retrieval		−	0.675

* Official or candidate-library classes; one fine-tune per context (Section D.4). GEARS predicted a candidate-invariant shift in every fit. Blank: not run.

context by five-fold cross-fitting (Section D.8) or kept as measurements from other cell lines; the held context never contributes a training response. Table 1 groups the solvers by the information they need; no optimization solver is instantiated. Profile retrieval ranks candidates by the cosine between the requested transition and their measured profile in the training contexts; CellNavi and GEARS are adapted under the same contract (Section D.4).

The K562 comparison uses the prespecified raw-target CED and $\overline{BU}$, fixed before the locked evaluation split was scored. Held-context comparisons use unit-direction targets, selected on these pools by a preregistered comparison, and $BU@20$, fixed before any external baseline was run; with raw targets the masked gain is +0.0430 [+0.0399, +0.0470] $\overline{BU}$, against +0.0470 with unit targets (Section E.1). Levels are medians over query-view rows. Gains are medians of paired differences, except for integer hit counts, where we report paired means; a gain need not equal a difference of levels. Intervals are 95% bootstrap intervals that resample query identities (locked split) or query genes (held contexts) with all their views, with the trained models held fixed (Section C).

5.2 VCDISIGN-CED IMPROVES INTERVENTION SELECTION

In held-out contexts, VCDesign-CED is the strongest solver we ran that can score without a measured response of the candidate (Figure 2a). With own responses masked, it improves on the identical base scorer by +0.053, +0.070 and +0.039 $BU@20$ in RPE1, HepG2 and Jurkat, and by +0.0492 [+0.0456, +0.0534] pooled, with 78.5% of rows improving (Figure 2b); among its first 20 selections it includes on average 1.11 more top-5% candidates than the base scorer (median +1). Used alone, the effect prediction, a linear embedding-to-response map (cf. Ahlmann-Eltze et al., 2025), reaches only 0.294. Goal alignment makes the gain hold in every held context: fused in the same way, the goal-blind norm of the unit-target ridge changes pooled BU by −0.0116 [−0.0180, −0.0062] against +0.0470 for CED, while a post hoc raw-target norm changes $\overline{BU}$ by +0.043, +0.068 and −0.079 in RPE1, HepG2 and Jurkat, against +0.048, +0.057 and +0.030 for raw-target CED (Section E.1). On RPE1 and Jurkat sub-pools with outcomes realized in held-out cell quarters, the share of the top 20 that realize an improvement (PHR@20) rises by a mean of +0.017 [+0.008, +0.026] but not in median, so the preregistered rule is null (Section D.4). The CellNavi adaptations, which need measured responses, stay below the base scorer in every held context; as single fine-tunes at the official step budget they describe these runs, not the method. In RPE1 the native run places the query’s own gene near chance (median rank 618 of 1,273) while reaching $BU@20$ 0.279, so identity recovery and outcome-based design can disagree.

On the locked K562 evaluation split, whose candidates neither branch has seen, adding CED raises median $\overline{BU}$ from 0.3551 to 0.3905 (Figure 2c). The median paired gain is +0.0319

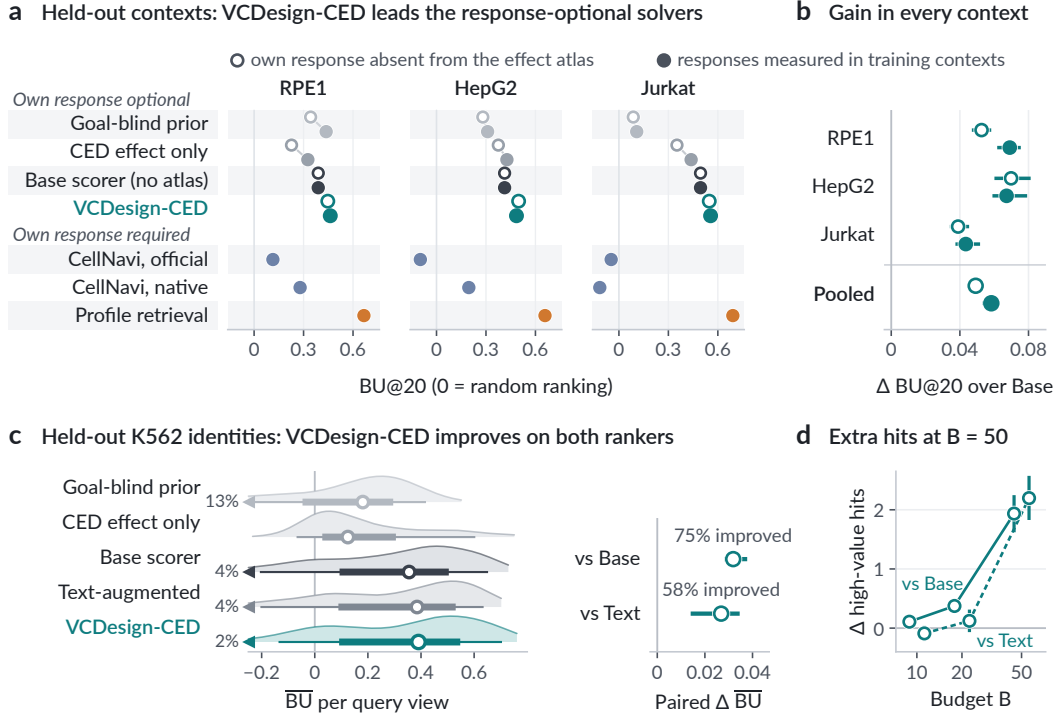


Figure 2: **VCDesign-CED improves selection in held-out contexts and on held-out identities.** **a**, Median BU@20 per held-out context (unit-direction CED); open: own response absent from the effect atlas, filled: measured in training contexts. **b**, Paired gain over the base scorer, 95% query-gene intervals. **c**, Locked K562 split (raw-target CED): row-level $\overline{BU}$ (density; bars: interquartile and 5th to 95th percentiles), axis cut at -0.25 with the share of rows below; right, median paired gains. **d**, Mean paired gain in high-value (top-5%) candidates among the first B . Intervals in **c**, **d**: 95% over query identities; intervals narrower than a marker are hidden.

$[-0.0282, +0.0378]$, with 75.1% of rows improving, and $+0.0268 [-0.0140, +0.0346]$ over the text-augmented ranker. The high-value-hit gain appears mainly at $B = 50$ (Figure 2d): among the first 50 candidates, VCDesign-CED selects on average 1.9 [1.6, 2.2] more high-value candidates than the base scorer and 2.2 [1.8, 2.6] more than the text-augmented ranker; at $B \leq 20$ it exceeds the base scorer by at most 0.4 and does not separate from the text-augmented ranker.

5.3 RESPONSE AVAILABILITY DEFINES THE SOLVER REGIME

When a candidate’s responses have been measured in other contexts, the best solver changes. Profile retrieval then reaches pooled BU@20 0.675 against 0.508 for VCDesign-CED, a paired advantage of $+0.1519 [+0.1424, +0.1602]$ (Figure 3a; Table 10). Measured responses should therefore be retrieved, and CED reserved for candidates without them.

What does CED transfer when the candidate’s own response is absent? The masking design changes one factor at a time (Figure 3b). Removing another masking fold, about 6% of atlas rows, leaves the gain unchanged: the pooled paired difference is $+0.0002 [-0.0001, +0.0005]$. Removing the candidates’ own responses at equal atlas size lowers it by 0.0078 [0.0061, 0.0096] in median (0.0154 in mean), most in RPE1 and not detectably in HepG2. Most of the gain is therefore carried by the responses of other candidates, possibly close interaction partners (Section 6).

Candidate libraries can mix both kinds of candidate. As the response-unseen share of a fixed pool rises from 0 to 100%, CED’s gain over the base scorer stays between $+0.039$ and $+0.070$ and changes by at most 0.016 within a context (Figure 3c). Measured candidates in these pools are still scored by CED; routing them to retrieval was not evaluated.

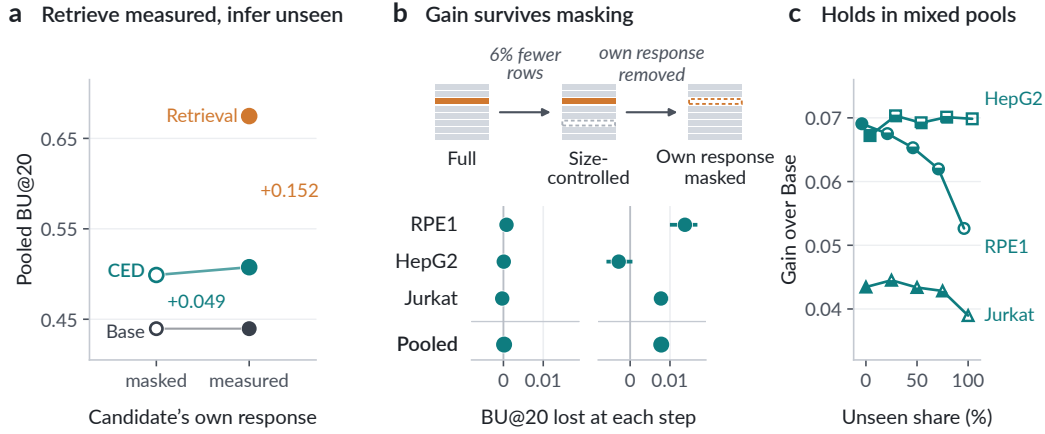


Figure 3: **Response availability sets the solver, and the response-unseen gain is transfer from other candidates' responses.** **a**, Pooled BU@20 with the candidate's own response masked from the effect atlas or measured elsewhere; labels are paired median gains; CED: VCDesign-CED. **b**, Paired BU@20 lost at each masking step, 95% query-gene intervals (most within the marker). **c**, Gain over the base scorer as 0 to 100% of each pool is made response-unseen (point estimates; fill: measured share).

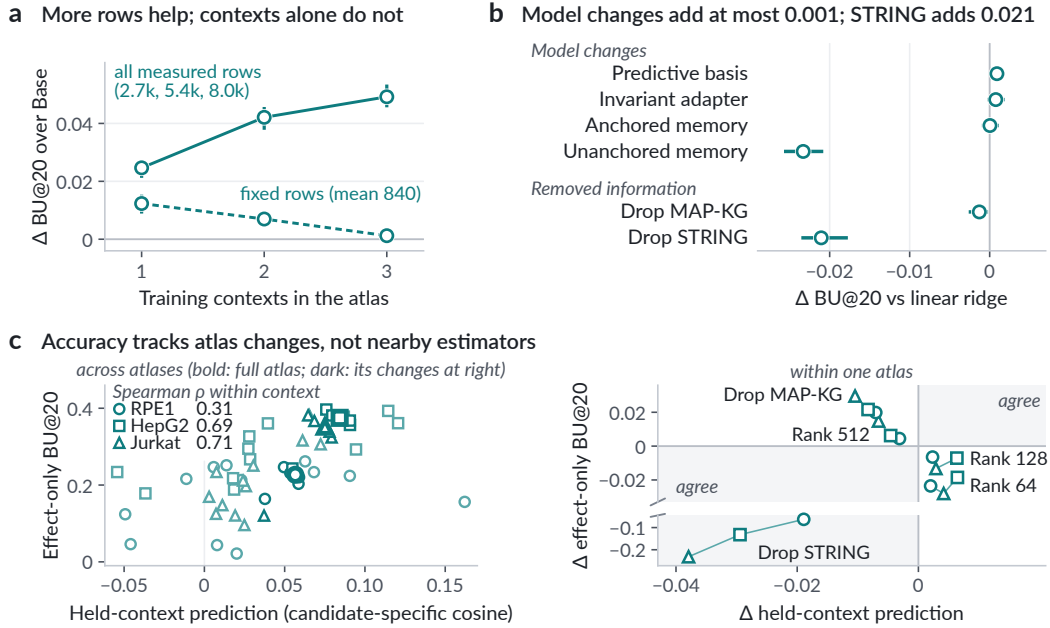


Figure 4: **Measurements and candidate information, not model changes, limit response-unseen design; prediction accuracy does not rank nearby estimators.** Response-unseen candidates, unit-direction ridge, BU@20. **a**, Gain over the base scorer with all rows (solid) or a fixed row count per held context, 840 on average (dashed). **b**, Change relative to the linear ridge, fused with the base scorer. Intervals in a, b: 95% over query genes, some within the marker. **c**, Effect-only BU@20 against held-context prediction, across 18 configurations per context and within one atlas (point estimates).

5.4 WHAT LIMITS RESPONSE-UNSEEN DESIGN?

Why is the masked gain modest? Controlled comparisons on masked candidates separate measurements, capacity and candidate information (Figure 4a,b). Across the tested controls, transfer tracks

measurement volume more consistently than context count. Atlases built from one, two and three training contexts with all their rows give pooled gains of +0.025, +0.042 and +0.049 over the base scorer. At a row count fixed per held context (840 on average), spreading the same rows over more contexts lowers the gain from +0.012 to +0.001 (Figure 4a), whereas unmasked raw-target arms rise from one context to two, then level off (Section D.10). Row matching also lowers the number of distinct identities, so this comparison separates measurement volume from context count but not from identity coverage. With identities held fixed and no masking, measuring each identity in all three contexts beats one measurement per identity spread over them by +0.028, +0.021 and +0.017 $\overline{\text{BU}}$ (raw targets; Appendix Figure 7).

The tested model changes add little, while candidate information matters. A predictive response basis, a context-invariant adapter and a nonlinear memory anchored to the ridge change BU@20 by at most +0.0009; an unanchored nonlinear memory loses 0.0233 [0.0208, 0.0257] (Figure 4b). Dropping STRING loses 0.0211 [0.0177, 0.0235], whereas dropping MAP-KG loses 0.0013. Within the tested model family, the gain is therefore bounded by the information and measurements the ridge receives rather than by its form; a factorial probe with a non-deployable measured-response reference agrees (Section D.3).

5.5 PREDICTION ACCURACY AND DESIGN QUALITY

The value of a better effect predictor depends on the decisions it changes. Across the masked atlas and input configurations, candidate-specific prediction accuracy and effect-only design utility are positively rank-correlated across held contexts, weakly in RPE1 (Spearman $\rho = 0.31, 0.69$ and 0.71 ; stratified permutation test over the three contexts, $p < 10^{-4}$; Figure 4c). Within a fixed atlas, however, four of five estimator changes move prediction and design in opposite directions in every held context: basis ranks 64 and 128 predict better but design worse, and rank 512 and dropping MAP-KG the reverse. Only dropping STRING lowers both. Prediction accuracy tracks large changes to the atlas, but choosing among nearby estimators requires evaluating the finite-budget ranking itself.

6 DISCUSSION AND CONCLUSION

The results set practical priorities. When candidates' responses have been measured, retrieval beats inference; without them, transfer from other candidates' responses adds value that grows with perturbation measurements and depends on protein-interaction knowledge, not on the effect-model capacity we tested. Because prediction accuracy did not order nearby estimators by design value, progress should be measured by the rankings a model produces.

Scope and Future Directions. The evidence covers single-gene CRISPRi in four cell lines with transcriptomic outcomes, graded on the 23.5 to 36.8% of usable identities whose transitions reproduce across replicate views. Directional BU@B is primary because these measurements support it reproducibly; realized-outcome guardrails are small and mixed (PHR@20 rises in mean but not median; the K562 realized-objective gain is +0.020 at $B = 10$ and slightly negative at larger budgets), so the gains are not yet confirmed as realized endpoint improvements. Three limits bound the response-unseen claim. No published method we ran ranks candidates with own responses masked, so that comparison is internal. Random gene folds leave most complex partners in the atlas, and STRING carries most of the knowledge-feature contribution, so part of the transfer may be to close partners; masking whole modules would test this. In a K562 stress test with 6,255 candidates absent from the other screens, where the base scorer was trained, the gain falls from +0.047 at 1,000 candidates to +0.006 at 3,000 and turns negative at 6,255 (−0.0103 [−0.0154, −0.0048] $\overline{\text{BU}}$ with raw targets, −0.0250 with unit targets; Appendix Figure 5). Genome-scale libraries, routing measured candidates to retrieval and context-aware maps (Section D.12) remain open.

AI USE STATEMENT

Generative AI tools were used to brainstorm and refine hypotheses and conceptual framing; provide feedback on methodology and experimental design; assist method implementation, data-reformatting code, and debugging; search and synthesize literature; interpret experimental results;

revise scientific figures; and draft and edit manuscript text. They were not used to generate synthetic datasets, formulate or prove mathematical claims, translate text, or conduct qualitative or thematic data analysis; proof-related tasks were not applicable. The authors checked AI-assisted ideas against the literature, reviewed and tested code, verified reported values against saved artifacts, and approved every scientific decision and claim. The authors take responsibility for the final content.

ETHICS STATEMENT

This work uses only publicly released CRISPRi Perturb-seq data from established human cell lines (K562 and RPE1 from Replogle et al. (2022); HepG2 and Jurkat from Nadig et al. (2025)) and public knowledge resources (Szklarczyk et al., 2023; Feng et al., 2026; Lin et al., 2023). No new experiments were performed, and no human-subjects or patient data were collected or used. VCDesign-CED ranks genetic interventions by how well they are predicted to realize a desired transcriptional transition in cultured cells. Its outputs are hypotheses that prioritize laboratory experiments, and they require experimental validation before any biological conclusion is drawn. The dual-use potential of such a ranking is limited, for three reasons. It ranks gene knockdowns, an intervention class already screened genome-wide in public atlases. It scores transcriptomic states of cell lines rather than organismal phenotypes. It creates no new biological agent or delivery method. Use in disease-related or therapeutic settings would call for functional validation and the biosafety and ethics review that such work normally requires.

REPRODUCIBILITY STATEMENT

Section 3 defines the design problem, the model interface and the evaluation interface, including the utility and the metrics, and Section A.1 gives their exact computation. Section 4 and Section B specify the base scorer, Candidate Effect Distillation and the fusion rule, with all hyperparameters, random seeds and compute. For Candidate Effect Distillation this covers the response basis, the features, the ridge penalty grid and how the penalty is selected. Section A describes the data sources, preprocessing, batch-disjoint replicate views, identity splits and the query-eligibility rule. Every split used in the paper (identity roles, replicate views and masking folds) is a deterministic SHA-256 hash of public identifiers, so it can be regenerated from the public data without stored index files. Section C lists the statistic, bootstrap procedure, seeds and resampling unit for every result family, and gives the SHA-256 hash of the configuration that was recorded before the held-out evaluation split was scored. Sections D and E report secondary endpoints and effect-model variants.

We will release anonymized code for data preprocessing, deterministic split and view construction, base-scorer training, Candidate Effect Distillation, fusion, evaluation and statistics, together with the configuration files of every experiment and the scripts that generate every table and figure. The release will include instructions to regenerate all processed atlases, candidate knowledge features and results from the public source data: the Perturb-seq screens of Replogle et al. (2022) and Nadig et al. (2025) (GEO GSE264667), STRING v12.0, and the released MAP-KG and ESM-2 assets.

DATA AND CODE AVAILABILITY

All experimental inputs are previously released datasets or public knowledge resources. K562 and RPE1 data come from the release associated with Replogle et al. (2022); HepG2 and Jurkat data are available from GEO under accession GSE264667 (Nadig et al., 2025). STRING v12.0, MAP-KG, and ESM-2 are available from their cited releases. Figure-level derived data are included as JSON and row-level CSV source files. The finalized external-baseline closure is additionally provided as a canonical JSON record with deterministic CSV views for levels, paired contrasts, PHR@20, compatibility, retention, and the prediction-design analysis. A source-data index and SHA-256 checksums accompany these files. The evaluation records immutable run-level manifests and independently witnessed score matrices. Anonymous code, configurations, manifests, and table-generation scripts will accompany the supplementary submission.

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A DATA AND METRICS

Contexts. Table 2 summarizes the four CRISPRi Perturb-seq screens. K562 and RPE1 come from Replogle et al. (2022). HepG2 and Jurkat come from the dual-sgRNA CRISPRi screens of Nadig et al. (2025) (GEO GSE264667, day 7, 10x Genomics 3' v3), read from the public single-cell H5AD files without reprocessing. In these two screens, identities are taken from the gene-level perturbation annotation, cells labelled non-targeting form the controls, and batches are the 10x GEM groups. All cells in the files are retained.

Two gene axes connect the contexts. K562 and RPE1 share 7,094 measured genes, 86.0% of the 8,248 K562 genes. The RPE1 zero-shot experiment places RPE1 states on the K562 gene axis at these shared genes. All four contexts share 6,550 genes. This shared 6,550-gene axis was fixed before any model was fit, and the leave-one-context-out, masking and atlas-scale experiments all use it.

HepG2 is shallower than Jurkat. Over all perturbed identities of each screen, including those below the usable depth, the median identity has 45 cells in HepG2 and 83 in Jurkat, summed over both views.

Table 2: **Per-context data.** K562 trains the model; RPE1, HepG2 and Jurkat are held out. Sources: K562 and RPE1, Replogle et al. (2022); HepG2 and Jurkat, Nadig et al. (2025). Usable identities have at least 25 cells in each of the two batch-disjoint replicate views. The shared axis is the 6,550-gene axis common to the four contexts. Eligible queries pass the measurability rule of Section A; for K562 they are counted on the held-out evaluation split. Candidate pools are listed per experiment, and a query’s own identity is always excluded from its pool. The three held-out contexts together contribute 1,333 queries (2,666 query-view rows) from 976 query genes to the pooled masking results. –: not applicable.

	K562	RPE1	HepG2	Jurkat
Role	training	held-out	held-out	held-out
Cells	1,989,578	247,914	145,473	262,956
Perturbations	9,866	2,393	2,393	2,393
Batches (views 1 / 2)	267 (131 / 136)	56 (26 / 30)	56 (26 / 30)	55 (26 / 29)
Measured genes	8,248	8,749	9,622	8,880
Usable identities	9,205	1,483	881	1,696
training and development identities on the shared axis	7,825			
Eligible queries	342 / 1,380	502	207	624
Pool, evaluation split	1,380	–	–	–
Pool, zero-shot	–	1,483	–	–
Pool, masking	1,430	1,273	764	1,450

Responses and queries. Counts are normalized to a library size of 10^4 and transformed by $\log(1+x)$. For an identity in view v , the goal state is the pseudobulk of the identity’s perturbed cells in the batches of v . The source state is a batch-matched control reference. It is the mean of the per-batch non-targeting control profiles, weighted by the identity’s cell counts across the batches of v . The transition is goal minus source. The consensus response $\mathbf{r}_c$, which CED uses as its target, is the mean of an identity’s two view transitions and is computed on training identities only. Every gene coordinate is kept. So when the knocked-down gene is on the measured gene axis, its own expression is part of the query transition and of every view response.

Batch-disjoint replicate views. A SHA-256 hash of the batch name assigns every batch of a context to view 1 or view 2 (K562: 131 and 136 batches). A second, independent hash splits each view into two quarters (K562: 66, 65, 63 and 73 batches), so the four quarters are also batch-disjoint. The same rule is applied to each context’s own batch names. K562 uses the quarters for the realized-objective analyses; RPE1, HepG2, and Jurkat additionally use them for the PHR guardrail on identities with at least 25 cells in every quarter (Section A.1).

K562 identity split. A SHA-256 hash of the identity name with a fixed salt assigns the 9,205 usable K562 identities to three splits: 6,445 training identities, a development split of 1,380 and a held-out evaluation split of 1,380 (the locked evaluation split of Section 5.1) ($6,445 + 1,380 + 1,380 = 9,205$). The splits are identity-disjoint. An evaluation-split identity is never a training target, a training query or a training candidate, and the split plays no part in checkpoint, penalty or fusion selection. Its configuration was recorded before it was scored (Section C).

Query eligibility. A transition can grade a ranking only if its two views agree, because one view forms the query and the other grades it. Consider a split of n identities with view responses $\mathbf{r}_g^{(1)}$ and $\mathbf{r}_g^{(2)}$ (Section 3). The agreement of query q is $E_q = \langle \mathbf{r}_{g_q}^{(1)}, \mathbf{r}_{g_q}^{(2)} \rangle$. Its one-sided permutation p-value against mismatched partners is

$$p_q = \frac{1 + |\{g' \neq g_q : \langle \mathbf{r}_{g_q}^{(1)}, \mathbf{r}_{g'}^{(2)} \rangle \geq E_q\}|}{n}. \quad (6)$$

A query is eligible when its Benjamini–Hochberg q-value (Benjamini & Hochberg, 1995) is at most 0.05 and $E_q > 0$. The rule and its level were fixed on the K562 training identities and are applied unchanged to every split and every context. On the evaluation split, 342 of 1,380 queries (24.78%) are eligible. In RPE1, HepG2 and Jurkat, 502 of 1,483 (33.9%), 207 of 881 (23.5%) and 624 of 1,696 (36.8%) usable identities are eligible. Eligibility filters queries only: on the evaluation split and in RPE1 zero-shot, ineligible identities remain in the candidate pool.

A.1 METRIC DEFINITIONS

Utility and ranking. The utility $U(q, c)$ is the opposite-view cosine of Equation (1). It is signed and continuous, and it is never a model input. It is computed from measurements independent of those that formed the query, because the grading view shares no batch with the query view. The pool $\mathcal{C}_q$ excludes the query’s own identity, and $N_q = |\mathcal{C}_q|$. A method’s ranking $\pi_q = (c_1, c_2, \dots)$ sorts $\mathcal{C}_q$ by descending score, with ties broken by ascending candidate identifier.

Budgeted utility. $\text{BU@}B$ and $\overline{\text{BU}}$ are defined in Equation (2) and Section 3.2, with rank weights $w_j = 1/\log_2(j + 1)$ and $\text{RU}^*\text{@}B$ the rank-weighted utility of the pool sorted by U . Two details complete the definition. First, the denominator of Equation (2) is floored at 10^{-8} , so a query whose best prefix equals the random expectation never divides by zero. Second, $\text{RU}_{\text{rand}}(q)$, the mean utility of the pool, is the exact expectation of $\text{RU@}B$ under a uniformly random ordering, because every rank then has the same expected utility. It therefore does not depend on B , and it is computed exactly, without sampled orderings.

Secondary metrics. $\text{MU@}B(q) = \frac{1}{B} \sum_{j=1}^B U(q, c_j)$ is the plain mean utility of the top B . $\text{HvHit@}B(q) = |\{j \leq B : U(q, c_j) \geq Q_{0.95}(q)\}|$ counts the top- B candidates at or above $Q_{0.95}(q)$, the 0.95 quantile of $\{U(q, c) : c \in \mathcal{C}_q\}$. Every metric is computed per query-view row, and tables report its median over rows.

Paired statistics and win rate. For a first and a second method evaluated on the same n rows, with per-row metric values $m_1(q)$ and $m_2(q)$, the paired difference of a row is $\Delta_q = m_1(q) - m_2(q)$. We report the median and the mean of Δ_q over rows with 95% bootstrap intervals (Section C), and the win rate $\frac{1}{n} \sum_q \mathbf{1}[\Delta_q > 0]$, the share of rows on which the first method is strictly higher.

Four-quarter realized outcomes. This secondary endpoint asks whether executing a top-ranked intervention moves the cell closer to the goal than doing nothing. The intervention may be attenuated. For an attenuation $a \in [0, 1]$,

$$J_{qc}(a) = 2a \alpha_{qc} - a^2 R_c, \quad (7)$$

which in expectation equals $\|\mathbf{d}_q\|^2 - \|a \mathbf{r}_c - \mathbf{d}_q\|^2$, the reduction in squared distance to the goal relative to no intervention. Here α_{qc} is the alignment of the candidate’s response with the query’s transition, and R_c is the candidate’s response energy. Both are computed as products across two batch-disjoint quarters. No measurement then multiplies itself, so neither statistic carries a noise-energy bias. For a query formed in view v , the attenuation a^* that maximizes J is chosen on the two quarters of v . The value $J_{qc}(a^*)$ is then realized on the two quarters of the opposite view.

Two summaries are reported. $J \text{RU@}B$ is $\text{RU@}B$ with the realized J in place of U , averaged over the two views of a query. $J>0@B$ is the share of top- B picks with positive realized J , pooled over queries in the first view. For the locked secondary endpoint, both are computed on the 1,053 evaluation-split identities with at least 25 cells in every quarter, which form the candidate pool, and on the eligible queries among them; the utility comparisons of Sections D.6 and E apply the same four-quarter construction to development-split queries. Values are reported in Section D.

The external-context PHR guardrail uses the same construction. For each query view, a^* is chosen on that view’s quarter pair and $J_{qc}(a^*)$ is realized on the opposite pair. The ranking is restricted to candidates with at least 25 cells in all four quarters, and the query must meet the same requirement. For the resulting top- B prefix,

$$\text{PHR@}B(q) = \frac{1}{B} \sum_{j=1}^B \mathbf{1}[J_{qc_j}(a_{qc_j}^*) > 0]. \quad (8)$$

Table 11 reports mean PHR@20 levels over query-view rows. Paired contrasts report both the mean and the preregistered median; pooled inference includes only contexts with at least 200 four-way candidates.

Latent endpoint value. The endpoint-distance objective compared in Sections D.6 and E is this same construction at full strength, $V_{qc} = J_{qc}(1) = 2\alpha_{qc} - R_c$: the estimated reduction in squared distance to the goal when the candidate’s response is realized undiscounted. It is used there as the utility in place of U , so a ranking formed on the two quarters of one view is graded by V on the two quarters of the other view. Unlike Equation (1) it keeps effect magnitude, and its second term $-R_c$ is a property of the candidate alone.

B IMPLEMENTATION DETAILS

This appendix lists every architectural choice and hyperparameter of VCDesign-CED, and Table 3 collects the settings in one place. Unless stated otherwise, all components operate on log-normalized pseudobulk states over the 8,248 genes of the K562 screen.

Base scorer architecture. The base scorer has 30,111,233 parameters. The source encoder maps x_s to the source embedding $\mathbf{h}_s$ through Linear(8248, 1024), LayerNorm, GELU, Dropout, Linear(1024, 512), LayerNorm and GELU. The transition encoder maps $\mathbf{d}_q = x_t - x_s$ to the transition embedding $\mathbf{h}_d$ through Linear(8248, 2048), LayerNorm, GELU, Dropout, Linear(2048, 1024), LayerNorm, GELU, Dropout, Linear(1024, 512) and LayerNorm. The FiLM layer (Perez et al., 2018) is one Linear(512, 1024) on $\mathbf{h}_s$, whose output is split into γ and β . Its weight is initialized to zero and its bias to $(\mathbf{1}, \mathbf{0})$, so the modulation is the identity at initialization. The query embedding is Linear(512, 256) applied to $\gamma \odot \mathbf{h}_d + \beta$, followed by ℓ_2 normalization.

On the candidate side, each knowledge modality with input dimension d has an adapter: LayerNorm(d), Linear(d , 512), GELU, Dropout, Linear(512, 128) and LayerNorm. A learned 128-dimensional modality embedding, initialized from $\mathcal{N}(0, 0.02^2)$, is added, and each token is ℓ_2 -normalized. Tokens are averaged over present modalities, projected by Linear(128, 256) and ℓ_2 -normalized. A candidate with no present modality receives the zero vector; there is no learnable placeholder token. All dropout rates are 0.10. The temperature is $\tau = \min(e^\theta, 100)$, with θ initialized to $\log(1/0.07)$.

Candidate knowledge. Both knowledge blocks come from public static sources. The STRING block is built from the STRING v12.0 physical subnetwork (Szklarczyk et al., 2023), using the experimental and database channels with text mining set to zero. It is compressed to 512 dimensions by truncated SVD (Halko et al., 2011) and standardized on the K562 training identities where present. The MAP-KG block is the 1,024-dimensional output of the released MAP-KG gene projector (Feng et al., 2026), applied to the released ESM-2 gene dictionary (Lin et al., 2023). The projector comes from the released knowledge-encoder checkpoint. That checkpoint is pretrained on 14 public static sources (gene, drug and relation databases), which contain no expression matrix or perturbation response. The separately released downstream perturbation-prediction checkpoint is not used.

On the development split, the STRING block covers 0.9232 of identities and the MAP-KG block 0.9333. Each block has a presence flag, which also serves as the pooling mask of the candidate encoder. 376 of the 6,445 K562 training identities have no modality. They lie outside the knowledge universe of both sources, so they score exactly zero in both branches.

Three further modalities are excluded from the knowledge set. Gene-description text embeddings can restate published perturbation phenotypes. A sequence embedding is redundant, because its ESM-2 asset is the dictionary that the MAP-KG projector already reads. Reactome pathway membership covers only 0.5964 of identities. Finally, STRING’s experimental channel and MAP-KG’s pretraining graph both include BioGRID relations. These features are identical in every compared arm.

Base scorer training. The base scorer learns from pairs of candidates whose utility gap exceeds the measurement noise. Each K562 training identity serves as a query in both replicate views, and the label is the utility of Equation (1), computed among training identities. For query-view row

q , let $\mathcal{S}_q$ be the union of three candidate sets: the utility top 128, the predicted top 128 under the current weights, and 256 uniformly random candidates. The query’s own identity is excluded. The measurement margin m_q is the median over candidates of $|\cos(\mathbf{r}_{g_q}^{(1)}, \mathbf{r}_c^{(1)}) - \cos(\mathbf{r}_{g_q}^{(2)}, \mathbf{r}_c^{(2)})|$. It is the typical disagreement between the utilities that the two replicate views assign to the same candidate. The pair set $\mathcal{P}_q$ holds all ordered pairs (c, c') in $\mathcal{S}_q$ with $U(q, c) - U(q, c') > m_q$, subsampled uniformly to at most 4,096 pairs. Over a minibatch $\mathcal{Q}$, the pairwise loss is

$$\mathcal{L}_{\text{pair}} = \frac{\sum_{q \in \mathcal{Q}} \sum_{(c, c') \in \mathcal{P}_q} \omega_{qcc'} \log(1 + e^{s_{\text{base}}(q, c') - s_{\text{base}}(q, c)})}{\sum_{q \in \mathcal{Q}} \sum_{(c, c') \in \mathcal{P}_q} \omega_{qcc'}}, \quad (9)$$

$$\omega_{qcc'} = \text{clip}\left(\frac{U(q, c) - U(q, c')}{\text{IQR}_q}, 0.25, 4\right) \rho(\text{rank}_q(c)),$$

where IQR_q is the interquartile range of $U(q, \cdot)$ over the candidate set. The rank weight ρ focuses the loss on the top of the ranking. It is 4, 2, 1 and 0.5 for a better candidate at utility rank at most 20, at most 64, at most 128, and beyond.

Training has two phases. Epochs 1 to 5 minimize $\mathcal{L}_{\text{pair}}$. Epochs 6 to 10 minimize $\mathcal{L}_{\text{pair}} + 0.5 \mathcal{L}_{\text{set}}$, where the set loss is

$$\mathcal{L}_{\text{set}}(q) = \log \sum_{c \in \mathcal{C}_q} e^{s_{\text{base}}(q, c)} - \log \sum_{c \in \mathcal{T}_q} e^{s_{\text{base}}(q, c)}, \quad (10)$$

averaged over the minibatch. It raises the set $\mathcal{T}_q$ of the 20 highest-utility candidates above the rest of $\mathcal{C}_q$. Its size matches the intended budget of about 20 experiments. Training uses AdamW (Loshchilov & Hutter, 2019) with learning rate 10^{-4} and weight decay 10^{-4} , gradient-norm clipping at 1.0, bfloat16 autocast, and seed 20260916. Each minibatch holds 8 training identities, that is, 16 query-view rows with both views of each identity. Checkpoints from epochs 6 to 10 are compared by median $\overline{\text{BU}}$ on the development split, and epoch 8 is selected. The fused model selects the same epoch under its own score, so VCDesign-CED and the base-scorer arm share every weight.

Text-augmented ranker. The text-augmented ranker has the dual-encoder architecture of the base scorer and is trained with the same recipe: seed 20260916, AdamW, the pairwise loss in epochs 1 to 5, and the set loss added in epochs 6 to 10. It differs in two ways. Its knowledge also includes gene-description text embeddings (NCBI gene descriptions encoded with all-MiniLM-L6-v2, 384 dimensions). A candidate without knowledge receives a learnable token. The base scorer omits the text, because the text can restate published phenotypes, and gives such candidates the zero embedding.

Candidate Effect Distillation. The consensus response is $\mathbf{r}_c = \frac{1}{2}(\mathbf{r}_c^{(1)} + \mathbf{r}_c^{(2)})$, the mean of the candidate’s two view transitions. The basis $\mathbf{W}$ is the rank-256 uncentered randomized SVD of the training consensus responses (seed 20260918, oversampling 32, four power iterations). The feature vector has 1,538 dimensions: the 512 STRING and 1,024 MAP-KG values, each multiplied by its presence flag, and the two flags. Features are standardized with the mean and standard deviation over training rows, and the scale is floored at 10^{-8} . The intercept is the mean target. The ridge is fit on the K562 training identities that carry at least one modality (376 of the 6,445 carry none). The penalty grid is $\lambda \in \{10^{-2}, 10^{-1}, \dots, 10^5\}$ (eight values). Five identity folds (fold seed 20260918) select the penalty with the highest mean held-out gene-space cosine between $\mathbf{W}\hat{\mathbf{z}}_c$ and $\mathbf{r}_c$; this is $\lambda = 10^4$ on K562. A candidate with no modality receives $\hat{\mathbf{e}}_c = \mathbf{0}$ and an effect score of 0.

Fusion. Nothing in the fusion is fitted. The per-query standardization z_q of Equation (5) uses the population standard deviation over the candidate set, floored at 10^{-12} , and the effect term enters with unit weight.

Cross-context deployment. The base scorer is the frozen K562 scorer in every context. States of another context are embedded on the 8,248-gene K562 axis at the shared genes, with zeros at the remaining genes. The effect predictor depends on the experiment. In the RPE1 zero-shot experiment, it is the K562 effect predictor as fitted on K562, and the effect cosine is computed on the 7,094 genes

Table 3: Hyperparameters of VCDesign-CED. The base scorer and CED are trained on the K562 training identities. Cross-validation on these identities selects the ridge penalty, and median BU on the development split selects the checkpoint epoch. Every value was fixed before the held-out evaluation split was scored.

Component	Setting	Value
Base scorer	parameters	30,111,233
	source / transition embedding width	512 / 512
	modality token width; joint embedding width	128; 256
	dropout	0.10
	temperature clamp; initial temperature	100; 1/0.07
Training	optimizer	AdamW
	learning rate; weight decay	10^{-4} ; 10^{-4}
	gradient-norm clip	1.0
	minibatch	8 identities (16 query-view rows)
	epochs (pairwise; pairwise + set)	1–5; 6–10
	pair pool: utility top; predicted top; random	128; 128; 256
	max. pairs per query	4,096
	gap-weight clip; rank weights	[0.25, 4]; 4/2/1/0.5
	set-loss top size; set-loss weight	20; 0.5
	selected epoch	8
	seed	20260916
CED	basis rank r	256
	SVD seed; oversampling; power iterations	20260918; 32; 4
	feature dimension	1,538
	penalty grid; CV folds	10^{-2} – 10^5 ; 5
	selected penalty (K562)	10^4
Fusion	effect weight; std. floor	1; 10^{-12}

that K562 and RPE1 share. In the leave-one-context-out, masking and atlas-scaling experiments, it is refit on the shared 6,550-gene axis of the four contexts. The refit uses one equally weighted row per (context, identity) pair, no context feature, and the rank, seed, penalty grid and cross-validation above. Per-fold refits under masking follow the protocol of Section D.8.

Compute. All experiments ran on one workstation with a single NVIDIA GeForce RTX 4090 GPU (24 GB) and an AMD Ryzen 9 7900X CPU, using Python 3.11, PyTorch 2.4.1 and NumPy 1.26. Training the base scorer takes 213 seconds for all ten epochs: about 19 seconds per pairwise epoch and 23 seconds per epoch with the set loss. CED is a closed-form ridge fit. The complete leave-one-context-out experiment, which refits CED 28 times (four held contexts, seven training atlases each) and scores and bootstraps every fold, took 4.3 minutes. The code release is described in the Reproducibility Statement.

C STATISTICAL PROCEDURES

Every comparison in the paper is paired. All arms rank the same candidate pool for the same query-view rows under the same utility, so each row yields one paired difference per contrast. This section states the replicate structure, the statistics and the resampling unit of each result family.

Replicate structure. Each usable identity gives two query-view rows. Each row forms its query in one view and grades it with the responses of the other (Section 3). A knockdown identity is usable when it has at least 25 cells in each of two batch-disjoint replicate views. Views are assigned by a SHA-256 hash of the batch name (K562: 131 and 136 batches), and the same rule is applied to each context’s own batch names. K562 identities carry about 108 cells per view. Over all perturbed identities, the median HepG2 and Jurkat identity has 45 and 83 cells summed over both views. A second hash splits each view into two quarters (K562: 66, 65, 63 and 73 batches). These quarters support the K562 realized objective and the external-context PHR guardrail reported in Section D.

Query eligibility. A query is graded only when its two replicate views reproduce its transition. The rule is a permutation test with Benjamini–Hochberg control, defined once in Section A. Eligible transitions are far more reproducible than rejected ones. On the development split, ranking candidates by their measured-response cosine in one view and grading in the other gives $\overline{\text{BU}}$ 0.7355 on eligible queries and 0.1571 on rejected ones.

Statistics. The level of an arm is its median over query-view rows. A contrast between two arms is the median of the per-row paired differences. On the held-out evaluation split, the mean paired difference is reported beside it (Table 5). The win rate is the share of rows with a strictly positive paired difference, so ties do not count as wins. Intervals are 95% percentile bootstrap intervals with 10,000 replicates and seed 20260916. In each replicate, resampling units are drawn with replacement, every row of a drawn unit enters, and the median (or mean) is recomputed. Table 4 lists the resampling unit of each result family. Intervals describe variation over queries with the trained models held fixed. The primary K562 comparison is VCDesign-CED against the base scorer on $\overline{\text{BU}}$; the primary cross-context contract comparison is the same contrast on masked BU@20. The K562 configuration was fixed before the locked evaluation split was scored; the cross-context configuration, whose unit-direction targets were selected on the same held-context pools by a preregistered comparison, was fixed before any external baseline was run. Other contrasts are secondary, and their intervals are not adjusted for multiplicity.

Prediction and design. The within-context association of Figure 4c is the Spearman coefficient over the distinct configurations of each held context (18 each, after removing three fits that duplicate another because the smallest atlas is already row-matched). Its significance is a stratified permutation test: design values are permuted within each held context, the statistic is the mean of the three within-context coefficients (0.572), and 20,000 permutations with seed 20260916 give $p < 10^{-4}$. The pooled coefficient over all 57 points, 0.610, treats points as independent and is reported only as a descriptive summary (Section D.4).

Row-level and query-identity intervals agree. K562 evaluation-split intervals follow the locked plan: a paired identity-cluster bootstrap that resamples both views of a query together. The originally recorded row-level intervals, which treat the two views as separate draws, agree closely (for example $[+0.0287, +0.0368]$ against $[+0.0282, +0.0378]$ for VCDesign-CED minus the base scorer, and $[1.69, 2.19]$ against $[1.62, 2.25]$ for its high-value-hit gain at $B = 50$); the only contrast whose interval changes side of zero is the secondary mean difference between the base scorer and the text-augmented ranker. For the value of historical response exposure, the two units give RPE1 $[+0.0045, +0.0094]$ against $[+0.0038, +0.0098]$, Jurkat $[+0.0054, +0.0088]$ against $[+0.0049, +0.0090]$, and K562 $[+0.0077, +0.0117]$ against $[+0.0075, +0.0123]$ (row-level first). Table 6 reports the query-identity intervals. The within-RPE1 comparison of effect atlases in Section D uses the row-level unit.

Seeds and deterministic components. The base scorer is trained once, with seed 20260916. VCDesign-CED and the base-scorer arm share its epoch-8 checkpoint, so the evaluation-split comparison isolates CED. The response basis is computed by randomized SVD (Halko et al., 2011) with seed 20260918 (32 oversampling columns, 4 power iterations). The ridge fit and the fusion rule are deterministic. The random-ranking baseline is a single draw. The raw-target row-matched atlas controls average five fixed row-sampling seeds per arm. The vocabulary-size curve takes the median over three random subsamples per size, and the full class is used at its own size. Every other assignment in the paper is a deterministic SHA-256 hash of an identifier: identity roles, replicate views, context quarters, masking folds, the response-unseen share of mixed vocabularies and the row subsamples of the unit-direction row-matched atlases.

Each CellNavi level comes from one fine-tune per context and class-set configuration, so no training-seed interval is claimed. GEARS uses the one measured-setting and five masked fits prescribed by the contract; every fit is degenerate. Intervals for external comparisons resample query genes with the trained models held fixed.

Evaluation-split configuration. The checkpoints, ridge penalty, fusion rule, eligibility rule and metrics of the evaluation-split comparison were recorded in one configuration file before

Table 4: **Resampling unit of each result family.** All intervals are 95% percentile bootstrap intervals (10,000 replicates, seed 20260916) of the median paired difference, or of the mean where a mean is reported. A cluster holds every query-view row that shares its label. Pooled rows cover RPE1, HepG2 and Jurkat, each held out from training in turn. Families that pool contexts resample query genes, so the views and contexts of one gene stay together.

Result family	Reported in	Rows	Resampling unit
K562 locked evaluation split	Table 5, Figure 2c,d	684 (342 queries)	query identity (both views)
RPE1 zero-shot, with fusion controls	Tables 7 and 14	1,004 (502 queries)	query-view row
Pooled masking gains	Table 6	2,666 (1,333 queries; 976 genes)	query gene (all views, contexts)
Pooled estimator ablations	Table 7; pooled rows of Tables 22 and 23	2,666 (1,333 queries; 976 genes)	query gene (all views, contexts)
External-baseline headline contrasts	Table 10, Figure 3a	pooled RPE1, HepG2 and Jurkat	query gene (all views, contexts)
Held-context gains over the base scorer, masking arm differences and estimator variants	Figure 2b, Figure 3b, Figure 4a,b	per context and pooled	query gene (all views, contexts)
High-value-hit differences	Figure 2d	684 (342 queries)	query identity (both views; mean)
PHR@20 contrasts	Table 11	four-way sub-pools; pooled inference uses powered contexts	query gene (all views, contexts)
Value of historical response exposure	Table 6	per held context	query identity (both views)
K562 fold at full class size (6,255 response-unseen, 1,570 measured)	Figure 5 and Tables 20 and 23	development split	query-view row
Development-split contrasts	Section D	development split	query-view row; query identity (views averaged) for the utility-choice intervals and the decision-aligned sampler, whose interval is on the mean

Point estimates (median paired difference, no interval): per-context raw-target masking gains; leave-one-context-out transfer; raw-target atlas-scale controls (Figures 6 and 7) (row-sampled arms averaged over five seeds); mixed vocabularies; vocabulary-size curve below full class size (median over three random subsamples per size); support terciles; reconstruction cosines; per-context rows of Tables 22 and 23 (raw norm alone, unit-direction targets, retention, win rate); the four-fold context-conditioned map and planted residual; CellNavi levels (one fine-tune per context and configuration).

the held-out evaluation split was scored, and the split was scored once with that configuration. The SHA-256 hash of that configuration is 720ee60edf8bf9ad0c35a5bbafc69ce31756f761a22a9a006f7d42366b55b1e5. The fusion form and its unit weight were chosen on the development split before this point.

D ADDITIONAL RESULTS

This appendix reports secondary endpoints of the held-out evaluation split, results on the development split, details of the context-transfer experiments, the controlled masking and vocabulary analyses of Section 5.3, the row-matched atlas arms and regression, the K562 fold of the held-context designs, and the context-specific structure of perturbation responses. Unless stated otherwise, entries are BU values or paired BU differences (median over query-view rows), and intervals are 95% bootstrap intervals with the resampling unit listed in Section C. Machine-readable source data for the external-baseline closure are included in the supplementary source package as one canonical JSON record and deterministic CSV views.

Table 5: **CED improves an identical base scorer on held-out K562 identities.** 342 queries (684 query-view rows); each row ranks 1,379 candidates. *Top:* median $\overline{\text{BU}}$, MU@B and HvHit@B over rows. Bold marks a unique best value among deployable rankers; the oracle reads measured responses and is a reference. *Bottom:* paired $\overline{\text{BU}}$ differences (first minus second method) with 95% paired identity-cluster bootstrap intervals (both views of a query resampled together, as the locked plan specifies) and win rates.

Method	$\overline{\text{BU}}$	MU@B		HvHit@B		
		$B=10$	$B=20$	10	20	50
Random	-0.0053	+0.0072	+0.0085	0	1	2
Goal-blind prior	+0.1807	+0.0344	+0.0421	2	5	10
CED effect only	+0.1245	+0.0279	+0.0256	1	2	5
Text-augmented ranker	+0.3850	+0.0767	+0.0642	4	7	10
Base scorer	+0.3551	+0.0754	+0.0629	4	6	9
VCDesign-CED	+0.3905	+0.0745	+0.0673	4	7	12
<i>Measured-response oracle (not deployable)</i>	+0.8142	+0.1458	+0.1247	9	16	30

Paired $\overline{\text{BU}}$ contrast	Median [95% CI]	Mean [95% CI]	Win rate
VCDesign-CED vs. base scorer	+0.0319 [+0.0282, +0.0378]	+0.0342 [+0.0274, +0.0414]	0.751
vs. text-augmented ranker	+0.0268 [+0.0140, +0.0346]	+0.0254 [+0.0150, +0.0360]	0.582
vs. goal-blind prior	+0.2272 [+0.2020, +0.2521]	+0.2243 [+0.2037, +0.2450]	0.836
Base scorer vs. goal-blind prior	+0.1912 [+0.1638, +0.2175]	+0.1900 [+0.1706, +0.2096]	0.811
vs. text-augmented ranker	-0.0066 [-0.0164, +0.0031]	-0.0088 [-0.0179, +0.0004]	0.465
CED effect only	+0.0659 [+0.0206, +0.1114]	+0.0845 [+0.0578, +0.1115]	0.576
vs. goal-blind prior			

Table 6: **CED keeps most of its gain for response-unseen candidates.** Paired $\overline{\text{BU}}$ differences of raw-target CED against the K562 base scorer, median over the query-view rows of each held context. Measured: full atlas. Response-unseen: each candidate’s responses removed from the atlas in every training context. Retained: response-unseen as a share of measured. Value of response exposure: size-controlled minus response-unseen arm at equal atlas size, with 95% bootstrap intervals over query identities. Pooled: RPE1, HepG2 and Jurkat together, with 95% bootstrap intervals over query genes. A candidate’s own response is worth a small fraction of the gain.

Held context (pool)	Measured	Response-unseen	Retained	Value of response exposure [95% CI]
RPE1 (1,273)	+0.0594	+0.0482	81%	+0.0063 [+0.0038, +0.0098]
HepG2 (764)	+0.0524	+0.0568	108%	-0.0025 [-0.0059, +0.0003]
Jurkat (1,450)	+0.0371	+0.0301	81%	+0.0069 [+0.0049, +0.0090]
Pooled	+0.0488	+0.0430		
	[+0.0456, +0.0526]	[+0.0399, +0.0470]		

D.1 COMPANION TABLES FOR MAIN RESULTS

Table 5 expands the held-out identity column of Table 1; Tables 6 and 7 are raw-target companions of the masking and ablation results. In this appendix, MASKED and SEEN denote the masked and measured settings of Table 1, BASE the base scorer, and the forward-then-match arm is the CED effect-only arm.

D.2 RESPONSE-AVAILABILITY CONTROLS

The unit-direction CED comparisons in Figure 3b separate each candidate’s own historical response from the overall size of the training atlas. The full-atlas, size-controlled, and response-unseen arms share queries, candidate pools, knowledge features, the fixed base scorer, and fusion. The size-controlled and response-unseen arms have equal atlas row counts; the held context contributes no training row in any arm. Their pooled median paired BU@20 gains over the base scorer are

Table 7: **Goal-conditioned direction carries the gain.** Median paired $\overline{BU}$ differences between each variant and its reference, with 95% bootstrap intervals. Upper block: terms fused with the base scorer in RPE1 zero-shot (1,004 query-view rows, resampled by query-view row). Lower block: unit-direction against raw-target CED, pooled over RPE1, HepG2 and Jurkat, each held out in turn (2,666 query-view rows, resampled by query gene). Win rate: share of rows with a strictly positive difference. Only the goal-conditioned effect score adds to the base scorer, and unit-direction targets refine it for measured and response-unseen candidates.

Variant	Reference	Setting	Median diff. [95% CI]	Win rate
VCDesign-CED	Base scorer	RPE1 zero-shot	+0.0196 [+0.0146, +0.0242]	0.629
Base + $\ \hat{e}_c\ $	Base scorer	RPE1 zero-shot	-0.0016 [-0.0104, +0.0142]	0.498
Base + random score	Base scorer	RPE1 zero-shot	-0.0511 [-0.0578, -0.0452]	0.252
Unit-direction targets	CED	Pooled, measured	+0.0048 [+0.0039, +0.0056]	-
Unit-direction targets	CED	Pooled, response-unseen	+0.0042 [+0.0032, +0.0051]	-

+0.0583 [+0.0530, +0.0625], +0.0584 [+0.0541, +0.0623], and +0.0492 [+0.0456, +0.0534], respectively.

Figure 3b plots the paired differences between adjacent arms, computed row by row from the same source rows with the registered query-gene bootstrap (10,000 replicates, seed 20260916); before computing them, the analysis script reproduces every arm interval above exactly. Full atlas minus size-controlled atlas (the cost of a smaller atlas) is +0.0008 [+0.0003, +0.0013] in RPE1, +0.0001 [-0.0002, +0.0004] in HepG2, -0.0003 [-0.0008, +0.0001] in Jurkat and +0.0002 [-0.0001, +0.0005] pooled. Size-controlled minus response-unseen (the value of the candidate’s own responses at equal atlas size) is +0.0138 [+0.0100, +0.0169], -0.0029 [-0.0059, +0.0006], +0.0077 [+0.0059, +0.0097] and +0.0078 [+0.0061, +0.0096]; its pooled mean is +0.0154 [+0.0127, +0.0182]. These are unit-direction $BU@20$ counterparts of the raw-target value of historical response exposure in Table 6.

In the mixed-pool experiment (Figure 3c), nested identity assignments make 0, 25, 50, 75, or 100% of candidates response-unseen while preserving the query and candidate pools. Gains remain positive at every share. The within-context range across shares is 0.0164 for RPE1, 0.0031 for HepG2, and 0.0055 for Jurkat. These unit-direction $BU@20$ results accompany the raw-target $\overline{BU}$ vocabulary analyses in Section D.9.

D.3 REPRESENTATION PROBE

A factorial probe on the K562 development split asks how much of the remaining gap lies in the candidate representation and how much in the query representation. One low-rank bilinear scorer (rank 128, 256-dimensional towers, a candidate-prior branch) is trained per cell with one fixed, lighter recipe than the main scorer (60 epochs, learning rate 10^{-4} , seed 20260916) and graded on 726 development query-view rows by median BU . The query is either the learned query embedding of the text-augmented ranker or the raw transition under a fixed Gaussian projection. The candidate is either its knowledge features (including text) or its own measured response in the query view, projected like the raw query; the latter is a non-deployable reference. The four cells give 0.1453 and 0.1331 with knowledge features and 0.6345 and 0.6365 with measured responses (learned and raw query, respectively). Replacing knowledge with measured responses therefore adds 0.489 and 0.503, whereas changing the query representation moves $\overline{BU}$ by at most 0.0121. Both query arms carry the full observed transition, so this probe shows that the learned query encoder loses little, and it bounds what better candidate information could add; it does not show that the query side carries no information. Its recipe underperforms the main scorer (deployable cells 0.133 to 0.145, against 0.344 for the main recipe), so its absolute levels are not comparable with Table 5.

D.4 EXTERNAL-BASELINE AND REALIZED-OUTCOME COMPANIONS

The external comparison follows one frozen evaluation contract across RPE1, HepG2, and Jurkat. CellNavi reproduces its released tutorial (accuracy 0.5101, weighted F1 0.4979 against 0.510 and 0.498) and is evaluated only in SEEN because its closed-set head has no class for a masked candidate (Wang et al., 2025). Each context uses one fine-tune from the released pretrained weights at the

official recipe (global batch 128, 1,050 steps, step-1000 checkpoint), on cells of the identities the SEEN effect atlas may use; the held context is never used for training. The official adaptation uses about 7,000 classes; the predeclared native-scale variant uses the candidate library itself (1,273, 764, and 1,450 classes), raising per-class exposure about six-fold without changing the trainer or step budget. Both versions learn: the final cross-entropy of the native version is 6.83, 6.16 and 6.86 against chance values of 7.15, 6.64 and 7.28. Matching the tutorial’s per-class exposure would have required about nine GPU-days per context, so these levels describe the gap to these runs, not what CellNavi could reach.

GEARS is legal in MASKED because it does not require a candidate’s measured response (Roohani et al., 2024). Its official reproduction matches the headline Top-20 differential-gene MSE (0.2456 against 0.2540) but fails the test-composition gate: the current package drops seven Norman conditions. Separately, all six contract fits predict a bitwise-identical shift for every candidate. We therefore retain GEARS as a labelled companion and use the internal forward-then-match arm as the informative forward comparison. PDGrapher is not run: only 45% of its PPI nodes lie on the shared gene axis, and its genetic mode requires a mutation vector absent for RPE1 (Gonzalez et al., 2025).

Table 8: **Compatibility and reproduction audit for published-method companions.** Contract status distinguishes whether a method can enter the response-unseen MASKED regime without changing its core model. The paper treatment follows the frozen compatibility and reproduction gates rather than the observed design score.

Method	Contract status	Reproduction outcome	Paper treatment
CellNavi	SEEN with interface adapter; no fair MASKED arm	Released tutorial passed; one fine-tune per context and class scale	SEEN companion only
GEARS	MASKED and SEEN with interface adapter	Headline MSE matched, but test composition failed; all six contract fits were candidate-invariant	Labelled degenerate companion
PDGrapher	Not directly compatible without a core change	Not run after the compatibility audit	Excluded from score tables

Table 9: **Unified external-baseline evaluation.** Entries are median BU@20 over query-view rows. MASKED is the response-unseen primary regime; SEEN allows a candidate’s response in the training contexts. The oracle reads held-context responses and is not deployable. –: incompatible with the regime or companion not reported as a level.

Method	MASKED	SEEN			
	pooled	RPE1	HepG2	Jurkat	pooled
Random	+0.002	−0.005	−0.010	−0.003	−0.005
Goal-blind prior	0.170	0.437	0.310	0.108	0.194
GEARS (degenerate companion)	–	−0.199	−0.100	−0.037	−0.078
CellNavi, official class set	–	0.114	−0.098	−0.047	−0.020
CellNavi, native class scale	–	0.279	0.196	−0.116	0.015
CED effect only (forward-then-match)	0.294	0.326	0.427	0.438	0.396
BASE	0.440	0.389	0.413	0.495	0.440
VCDesign-CED	0.499	0.462	0.485	0.556	0.508
Profile retrieval	–	0.666	0.658	0.691	0.675
<i>Measured-response oracle</i>	<i>0.951</i>	<i>0.964</i>	<i>0.937</i>	<i>0.938</i>	<i>0.951</i>

Native-scale CellNavi improves over its official adaptation in RPE1 and HepG2 but not Jurkat. Its pooled BU@20 remains below BASE. These rows are descriptive point estimates from one fine-tune per context and configuration. The query’s own class has median rank 618 of 1,273 in RPE1, 238 of 764 in HepG2, and 816 of 1,450 in Jurkat, even when other candidates achieve useful design scores. Identity recovery and outcome-based design therefore measure different orderings.

Table 10: **Paired headline contrasts under the unified contract.** Values are median per-row $\Delta\text{BU}@20$ with 95% query-gene bootstrap intervals. GEARS-ridge compares estimators on the same K562 atlas and masks.

Regime	Contrast	Pooled difference [95% CI]
MASKED	VCDesign-CED – BASE	+0.0492 [+0.0456, +0.0534]
MASKED	VCDesign-CED – GEARS	+0.5531 [+0.5315, +0.5709]
MASKED	GEARS – unit ridge, same atlas	−0.3529 [−0.3735, −0.3330]
SEEN	VCDesign-CED – CellNavi, native scale	+0.3363 [+0.3025, +0.3652]
SEEN	VCDesign-CED – Profile retrieval	−0.1519 [−0.1602, −0.1424]

Among the first 20 masked selections, VCDesign-CED includes on average 1.11 more high-value candidates than BASE (median +1; companion contrasts of the external closure).

Four-quarter realized-outcome guardrail. $\text{PHR}@B$ is the fraction of the selected prefix with independently realized positive endpoint improvement (Section A.1). It is available only when the query and candidate have at least 25 cells in each of four batch-disjoint quarters. RPE1 and Jurkat pass the preregistered minimum of 200 candidates; HepG2 is reported but underpowered.

Table 11: **PHR@20 on the four-way sub-pool.** BASE and VCDesign-CED use MASKED rankings. The oracle is not deployable. HepG2 is below the preregistered 200-candidate threshold and is excluded from the pooled powered contrast.

Context	Candidates	Queries	Powered	BASE	VCDesign-CED	Oracle
RPE1	247	86	yes	0.477	0.503	0.787
HepG2	74	16	no	0.648	0.670	0.798
Jurkat	342	135	yes	0.692	0.703	0.835

Across the powered contexts, the VCDesign-CED minus BASE contrast has mean +0.0171 [+0.0083, +0.0258] and median 0.000. $\text{PHR}@20$ changes in steps of 0.05, so most rows tie and the preregistered median rule returns NULL. The oracle contrast is +0.208 [+0.186, +0.232], showing that the endpoint itself has headroom.

Prediction and design. Within each held context, held-context prediction quality and effect-only design are rank-correlated over the 18 distinct atlas and input configurations (Spearman 0.31, 0.69 and 0.71 in RPE1, HepG2 and Jurkat; stratified permutation $p < 10^{-4}$, Section C). Pooled over all 57 fits, which include three duplicates, the coefficient is 0.610 (0.645 in SEEN), a descriptive value that treats fits as independent. Within a fixed atlas, prediction and design changes agree in sign for 3 of 15 MASKED variant-context cells (9 of 15 in SEEN): ranks 64 and 128 predict better but design worse, and rank 512 and dropping MAP-KG predict worse but design better, in every held context. Prediction quality detects large distribution shifts but does not select among nearby design estimators.

D.5 HELD-OUT EVALUATION SPLIT: SECONDARY ENDPOINTS

Table 12 adds the four-quarter realized objective (Section A.1) to the evaluation-split comparison of Table 5. It is computed on the 1,053 candidates and the eligible queries measured in all four batch-disjoint quarters, so its population is a subset of the main table’s. VCDesign-CED raises the realized-objective $\text{RU}@10$ over the base scorer by +0.0201 (difference of medians) and raises the share of top-10 picks with a positive realized objective from 0.501 to 0.513; the text-augmented ranker is highest among deployable rankers on both quantities. Deeper in the budget the realized-objective difference is slightly negative (−0.0049 at $B = 20$, −0.0053 at $B = 50$), while the share of positive picks still rises at $B = 20$. Paired mean utility of the top 20 rises by a median of +0.0043 (mean +0.0066); for the top 10 the median paired change is −0.0009 and the mean +0.0052. The share of top-10 picks that carry no knowledge modality falls from 0.0035 to 0.0016.

Table 12: **Four-quarter realized objective on the held-out evaluation split.** Median realized-objective RU@10 and the share of top-10 picks with a positive realized objective, computed on the 1,053 candidates and the eligible queries measured in all four quarters.

Method	Realized RU@10	Share with $J > 0$ in top 10
Random	-0.03145	0.269
Goal-blind prior	+0.01529	0.374
CED effect only	+0.01029	0.383
Text-augmented ranker	+0.08307	0.517
Base scorer	+0.05286	0.501
VCDesign-CED	+0.07296	0.513
<i>Measured-response oracle (not deployable)</i>	+0.36692	0.732

D.6 DEVELOPMENT SPLIT

The development split selected the checkpoint (both arms at epoch 8) and the fusion form. Table 13 shows that the evaluation-split pattern is already present there. Adding CED to the base scorer raises $\overline{BU}$ from +0.2928 to +0.3328 at the selected epoch. On the text-augmented ranker, CED adds +0.0576 [+0.0517, +0.0636] with a win rate of 0.807, whereas fusing the goal-blind predicted magnitude (win rate 0.383) or random scores (win rate 0.176) through the same rule lowers $\overline{BU}$. The standalone effect score is 93% query-conditional: its query-independent share is 0.0712. On the development split, adding CED with the unit weight to the text-augmented ranker gives +0.3734 (the fitted weight below reaches +0.3907); the evaluation-split comparison uses the text-free base scorer.

Table 13: **Development split.** Upper: $\overline{BU}$ levels. Lower: paired $\overline{BU}$ differences against the stated base, with 95% bootstrap intervals over query-view rows. The fusion ablations use the text-augmented ranker as the base; their win rates are 0.807 (CED), 0.383 (goal-blind magnitude) and 0.176 (random score).

Base	Arm	$\overline{BU}$
Base scorer	base alone	+0.2928
Base scorer	+ CED (VCDesign-CED)	+0.3328
Text-augmented ranker	base alone	+0.3444
Text-augmented ranker	+ CED	+0.3734
Base	Fused term	Paired difference
Text-augmented ranker	CED	+0.0576 [+0.0517, +0.0636]
Text-augmented ranker	goal-blind magnitude	-0.0568 [-0.0684, -0.0338]
Text-augmented ranker	random score	-0.1147 [-0.1248, -0.1060]

Fusion form and weight. On the development split, with the text-augmented ranker as the base, the unit-weight z-sum adds +0.0576 [+0.0517, +0.0636] (win rate 0.807) and a rank-average fusion adds +0.0199. Weighting the effect score by 0.25, a weight fitted on K562 training identities, adds +0.0344 [+0.0315, +0.0385] (win rate 0.864) and reaches a median level of +0.3907, against +0.3734 for the unit weight. We use the unit weight: it involves no fitting, and on the development split it also raises the top-10 mean utility and the realized-objective RU@10 over the text-augmented ranker, which the fitted weight leaves flat.

Candidate-side probe. The factorial probe of Section D.3 also moves uniform pair accuracy from 0.5228 and 0.5245 with knowledge features to 0.5871 and 0.5883 with measured responses, whereas changing the query representation moves it by +0.002 and +0.001. In a separate experiment, scorer capacity is not limiting either: a trilinear scorer reaches uniform pair accuracy 0.5261 against 0.5252 for the bilinear form used by the base scorer. Sampling training pairs aligned with the budget decision changes $\overline{BU}$ by a median of -0.0794 (mean -0.0820 [-0.0920, -0.0720], bootstrap over query identities with the two views averaged), so the top-focused pairwise loss of Section B is kept.

Replicate ceiling. On the development split, ranking candidates by one replicate view of the utility and grading with the other gives the level reached by ranking with a single replicate measurement of every candidate’s response: BU@10 0.766 and 0.797, BU@20 0.736 and 0.738, BU@50 0.687 and 0.698 for the two grading directions. It is computed on a different population (development split, candidates measured in all four quarters) from the evaluation-split measured-response oracle (+0.8142), so the two are not directly comparable.

Choice of utility and score form. The directional utility introduced in Section 3.2 was selected through comparisons with four alternatives to the utility or response representation. This paragraph gives their construction and evaluation populations; Table 23 reports the values. All four rank candidates from measured responses on development-split queries, restricted to the identities measured in all four batch-disjoint quarters; no model is trained and no model score enters any value. The latent endpoint value replaces the direction cosine as the utility, so a ranking formed on one pair of quarters is graded by that same quantity on the other pair (Section A.1). The attenuation-aware ranking scores a candidate by the realized $J_{qc}(a^*)$ of Section A.1 instead of by direction, and it is graded, like the two distributional variants, by the realized objective rather than by U . The kernel (RBF) distributional transition replaces each pseudobulk transition by a kernel mean embedding of the identity’s single-cell population, computed in a frozen low-dimensional projection of the gene axis, which is what makes a nonlinear kernel tractable. The tie-breaking variant keeps the pseudobulk ordering and reorders only within a pre-declared score tolerance, with the distributional score supplied directly rather than predicted, so it bounds what tie-breaking could contribute. Intervals over these contrasts resample query identities with the two grading directions averaged, and they are given on the mean (Section C).

D.7 CONTEXT TRANSFER DETAILS

In the RPE1 zero-shot experiment (Table 14, first row), the mean utility of the top 10 rises from +0.2426 (frozen base scorer) to +0.2682 (VCDesign-CED) and that of the top 20 from +0.2163 to +0.2437; the measured-response oracle reaches BU +0.9606 on the same pool. The frozen base scorer alone already exceeds random ranking by +0.3085 [+0.2811, +0.3309] with a win rate of 0.843, so the effect branch improves a ranker that is already useful in RPE1. The split of RPE1 queries by whether their knockdown is a K562 training identity (313 and 189 queries) separates queries only: both groups rank the same 1,483 candidates. The zero-shot experiment uses the frozen K562 map on the K562 gene axis, whereas the leave-one-context-out experiments refit the map on the shared 6,550-gene axis, so the two K562-only values for RPE1 (+0.0196 and +0.0201) come from different maps on different gene axes.

Leave-one-context-out atlases. Table 14 gives every leave-one-context-out atlas and the zero-shot entry. Which contexts enter the atlas matters. A Jurkat-only atlas gives +0.0488 on held RPE1, more than twice the K562-only gain of +0.0201. A HepG2-only atlas contributes little (+0.0057 on RPE1, +0.0012 on Jurkat); HepG2 has 881 usable identities and its median perturbed identity 45 cells (both views, all identities), against 1,696 and 83 for Jurkat. The K562 column is discussed in Section D.11.

Within-RPE1 effect atlases. A second RPE1 analysis asks which effect atlas serves RPE1 best when RPE1 responses may enter training. Three CED maps are fit on the 7,094 genes shared by K562 and RPE1, with no context feature: K562 only, RPE1 only, and both pooled. Predictions for candidates inside a map’s own training pool are out of fold, so no candidate’s own RPE1 response fits its own prediction; because other RPE1 candidates do enter training, this is not a held-context design. Table 15 shows that pooling the two contexts more than triples the gain of the K562-only map and beats the RPE1-only map, in line with the scaling result of Section 5.4.

D.8 CONTROLLED MASKING DETAILS

Masking protocol. A SHA-256 hash of the gene symbol assigns every candidate to one of five folds. For each fold, the response-unseen arm removes the fold’s responses from the distillation atlas in every training context, recomputes the response basis $\mathbf{W}$ from the remaining rows, and refits the ridge by the recipe of Section B to predict effects for the fold. Cross-fitting over the five folds makes

Table 14: **Context transfer.** $\overline{BU}$ differences of VCDesign-CED against the K562 base scorer (median over query-view rows of the held context). Zero-shot: the K562 map on the K562 gene axis, nothing trained on RPE1, 95% bootstrap interval over query-view rows. Leave-one-context-out: the held context contributes no training row, CED is refit on the shared 6,550-gene axis, point estimates. $-$: the atlas contains the held context; empty: not reported. † Base scorer trained on K562; development-split queries.

Effect atlas	Held RPE1	Held HepG2	Held Jurkat	Held K562 †
<i>Zero-shot, K562 gene axis</i>				
K562 map, unchanged	+0.0196 [+0.0146, +0.0242]			$-$
<i>Leave one context out, shared 6,550-gene axis</i>				
K562 only	+0.0201	+0.0318	+0.0277	$-$
Jurkat only	+0.0488		$-$	+0.0071
RPE1 only	$-$		+0.0259	-0.0021
HepG2 only	+0.0057	$-$	+0.0012	-0.0015
Three other contexts	+0.0594	+0.0518	+0.0343	+0.0076
Pool (usable identities)	1,483	881	1,696	7,825

Table 15: **Within-RPE1 effect atlases.** Paired $\overline{BU}$ differences against the frozen K562 base scorer on the RPE1 zero-shot queries and pool (Table 14), with 95% bootstrap intervals over query-view rows. The pooled atlas has a win rate of 0.880.

Effect atlas	Paired difference
K562 only	+0.0182 [+0.0137, +0.0235]
RPE1 only (out of fold)	+0.0287 [+0.0259, +0.0320]
K562 and RPE1 pooled (out of fold)	+0.0657 [+0.0606, +0.0698]
Pooled minus K562 only	+0.0366 [+0.0332, +0.0401]

every candidate in the pool response-unseen. Removing a fold also shrinks the atlas, so the size-controlled arm scores each fold with the next fold’s masked predictor: its atlas is reduced by the same amount, but the evaluated candidates stay measured. The measured arm uses the full atlas. All three arms share the knowledge features k_c , the pool, the queries, the utility, the fusion rule and the metric, so the size-controlled and response-unseen arms differ only in whether a candidate’s own responses supervised its predicted effect. A held context’s pool holds its usable candidates that some training context measured and that carry at least one knowledge modality. All arms also share the K562 base scorer, whose training identities include about two thirds of the masked candidates, so every masking contrast measures the effect branch.

Per-fold value of historical response exposure. Table 6 cross-fits the five masking folds into one pool per context. Table 16 instead scores each fold on its own masked candidates, which gives pools of 225 to 304 candidates in RPE1 and Jurkat and 138 to 175 in HepG2. Every fold has a positive value of historical response exposure in RPE1 and Jurkat, and HepG2 folds lie between -0.0003 and $+0.0096$. In RPE1, HepG2 and Jurkat the per-fold medians exceed the cross-fitted values; in K562 the per-fold median ($+0.0081$) is below the cross-fitted value ($+0.0097$). Because $\overline{BU}$ differences depend on pool size (Section D.9), Table 6 uses the cross-fitted values, which rank each context’s full pool.

Atlas shrinkage. Masking also removes rows from the atlas. The cost of this shrinkage alone, the measured minus the size-controlled arm, is $+0.0031$ (RPE1), -0.0003 (HepG2), -0.0000 (Jurkat) and $+0.0003$ (K562, development-split queries) on the cross-fitted pools of Table 6 and Table 21. Medians of paired differences are not additive, so measured minus response-unseen need not equal the value of historical response exposure plus shrinkage.

Comparability of the fold predictors. Cross-fitting mixes the five fold predictors’ per-candidate columns into one score row, so the fold predictors should shrink alike: select the same ridge penalty, and give the same share of each prediction to the candidate-specific part rather than the common

Table 16: **Value of historical response exposure per masking fold.** Size-controlled arm minus response-unseen arm, each fold scored on its own masked candidates. [†]Base scorer trained on K562; development-split queries.

Held context	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Median
RPE1	+0.0032	+0.0255	+0.0189	+0.0031	+0.0123	+0.0123
HepG2	+0.0096	+0.0001	-0.0003	+0.0019	+0.0080	+0.0019
Jurkat	+0.0103	+0.0156	+0.0056	+0.0057	+0.0158	+0.0103
K562 [†]	+0.0081	+0.0073	+0.0029	+0.0148	+0.0184	+0.0081

response. In RPE1, HepG2 and Jurkat the full-atlas predictor and all five fold predictors select the same penalty, and their candidate-specific shares differ by factors of at most 1.041, 1.031 and 1.068 (1.068 also among Jurkat’s five fold predictors alone); Jurkat’s measured-versus-response-unseen share (81%) therefore mixes fold predictors that shrink slightly differently. In K562, the full-atlas predictor of the measured arm selects a penalty of 10^3 against 10^4 in all five fold predictors, and its candidate-specific share is 1.792 times theirs. K562’s measured-versus-response-unseen contrast, including its 4% retained share, therefore compares unlike predictors. With unit-direction targets (Sections 4.1 and E.1), the shares agree within factors of 1.042, 1.046 and 1.028, and response-unseen candidates retain 0.76, 1.04 and 0.90 of the measured BU@20 gain in RPE1, HepG2 and Jurkat (Table 22). The value of historical response exposure does not depend on this comparability: the size-controlled and response-unseen arms are built from the same five fold predictors, permuted by one fold, so any mixing distortion enters both arms alike and cancels in their difference.

Within-context reference. When the held context is also a training context, the measured arm is fit on the evaluated candidate’s own response in the very context that grades it, so its value of historical response exposure is an upper bound. Fold medians of this value are then +0.0218 (RPE1), +0.0228 (HepG2), +0.0230 (Jurkat) and +0.0200 (K562, development-split queries), against the cross-context values of Table 6. Response-unseen candidates still beat the base scorer in RPE1, HepG2 and Jurkat in this design (+0.0588, +0.0215, +0.0279). The cross-context design avoids this self-supervision and is used in Section 5.3.

Static support. Table 17 splits response-unseen candidates into equal-count terciles of ridge leverage against the full training atlas, with identical strata in both arms. The gain over the base scorer is monotone in all three held contexts and largest where static support is weakest. We interpret leverage here as distinctiveness: a candidate near the centre of the training atlas has a predicted effect close to the common response, which the effect score can hardly separate from its neighbours, whereas a distant candidate has a distinctive predicted effect. Terciles are point estimates; the ordering is the same in all three contexts.

Table 17: **Response-unseen gain by static support.** Paired $\overline{BU}$ differences against the frozen K562 base scorer (point estimates) within terciles of ridge leverage.

Held context	High support	Medium support	Low support
RPE1	+0.0119	+0.0328	+0.0630
HepG2	+0.0205	+0.0338	+0.0487
Jurkat	-0.0113	+0.0170	+0.0395

Reconstruction accuracy and design value. Table 18 compares the median cosine between predicted and measured consensus response, $\cos(\hat{e}_c, r_c)$, with the design gain retained under masking. Masking costs 14 to 21% of reconstruction accuracy but 0 to 19% of the design gain in RPE1, HepG2 and Jurkat; in HepG2 it costs 16% of accuracy and none of the design gain. Ranking uses the direction of a predicted effect relative to the other candidates, which is more robust to masking than its absolute accuracy.

Table 18: **Reconstruction accuracy and design value under masking.** Median $\cos(\hat{\mathbf{e}}_c, \mathbf{r}_c)$ on the evaluated candidates, its relative drop under masking, and the share of the measured design gain that response-unseen candidates retain (Table 6). [†]Base scorer trained on K562; development-split queries. [‡]Measured and response-unseen predictors differ in penalty (Section D.8). [§]Jurkat’s fold predictors differ in candidate-specific share by a factor of 1.068 (Section D.8).

Held context	Measured	Response-unseen	Accuracy drop	Design gain retained
RPE1	0.2981	0.2360	21%	81%
HepG2	0.3738	0.3155	16%	108%
Jurkat	0.2158	0.1847	14%	81% [§]
K562 [†]	0.2635	0.2309	12%	4% [‡]

D.9 MIXED VOCABULARIES AND VOCABULARY SIZE

In the mixed vocabularies of Section 5.3, a nested hash of the gene symbol makes 25, 50 or 75% of each masking pool response-unseen, with the same queries; these candidates take their predicted effect from the masking predictor that excluded them, and the rest keep the measured arm’s prediction. Figure 3c applies this design to the unit-direction estimator and BU@20. Table 19 is a raw-target, $\overline{\text{BU}}$ companion that analyzes promotions at a 50% response-unseen share. A promotion is a candidate that enters the top B of a query-view row under the fused score but not under the base scorer alone; its entrant value in that row is its utility minus the mean utility of the candidates it displaced. Each promoted candidate’s entrant values are averaged over its promotions (both views, all queries), and Table 19 summarizes over promoted candidates. Promoted candidates are worth more than the candidates they displace in every context, with mean entrant values of +0.0416 to +0.0629 at $B = 10$ and +0.0236 to +0.0317 at $B = 50$. Response-unseen entrants are worth 84 to 107% of measured entrants at $B = 10$; at $B = 50$ this falls to 63% and 77% in RPE1 and Jurkat, so the cost of a response-unseen vocabulary sits in the deeper part of the budget rather than in the first picks.

Table 19: **Entrant value at a 50% response-unseen share.** Promoted: number of distinct candidates promoted at least once. Entrant value: mean over promoted candidates of each candidate’s average entrant value, overall and by class. Positive: share of promoted candidates whose average entrant value is positive.

Held context	B	Promoted	Mean	Measured	Response-unseen	Positive
RPE1	10	382	+0.0616	+0.0666	+0.0562	0.688
RPE1	50	987	+0.0317	+0.0390	+0.0244	0.638
HepG2	10	251	+0.0629	+0.0608	+0.0650	0.661
HepG2	50	594	+0.0316	+0.0306	+0.0327	0.621
Jurkat	10	446	+0.0416	+0.0415	+0.0418	0.614
Jurkat	50	1,078	+0.0236	+0.0266	+0.0204	0.612

Figure 5 and Table 20 report the separate raw-target vocabulary-size stress test for the response-unseen class, the measured class and all candidates of the K562 fold, whose queries are development-split identities and whose 7,825 candidates are K562 training and development identities. The three subsamples disagree by more than the effect at 250 and 500 candidates for the response-unseen class and for all candidates, and at 250 to 1,000 for the measured class (vertical ranges in Figure 5); at 6,255 its interval is $[-0.0154, -0.0048]$, and the full measured class (1,570) gives +0.0095 $[+0.0066, +0.0120]$. At matched size the response-unseen class gains more than the measured class (+0.0472 against +0.0023 at 1,000 candidates, +0.0198 against +0.0095 at 1,570), so the decline with vocabulary size is not a cost of missing responses.

D.10 ATLAS SCALE: ROW-MATCHED ARMS AND REGRESSION

Figure 6 shows an unmasked, raw-target version of the row-matched comparison ($\overline{\text{BU}}$; evaluated candidates keep their own training-context rows). At a matched number of rows, the gain over the RPE1, HepG2 and Jurkat folds, drawn as the blue line, is +0.01679 with one training context, +0.02263 with two and +0.02190 with three; over all four folds, including the K562 fold, it

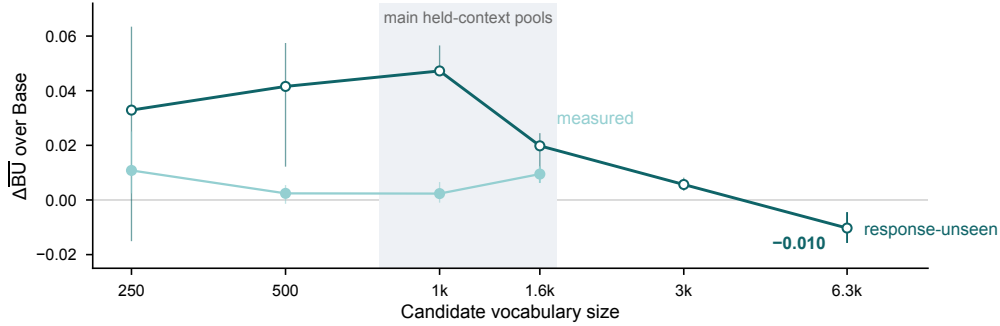


Figure 5: **Vocabulary-scale stress test on the K562 development fold.** Paired $\Delta\overline{BU}$ over the frozen K562 base scorer for raw-target CED as measured or response-unseen candidate vocabularies grow. Points below the complete class size are medians over three deterministic subsamples and vertical lines span the three values; complete-class points show 95% row-bootstrap intervals. The shaded range contains the candidate-pool sizes of the main held-context experiments. The response-unseen gain is -0.0103 $[-0.0154, -0.0048]$ at 6,255 candidates, so genome-scale vocabularies remain outside the regime established by Figure 3.

Table 20: **CED gain against vocabulary size (K562 fold, development-split queries).** Paired $\overline{BU}$ differences against the frozen K562 base scorer, median over three random subsamples per size (the full class at its own size). $-$: size exceeds the class. *The three subsamples disagree by more than the effect.

Class	250	500	1,000	1,570	3,000	6,255
Response-unseen	+0.0329*	+0.0416*	+0.0472	+0.0198	+0.0057	-0.0103
Measured	+0.0108*	+0.0024*	+0.0023*	+0.0095	-	-
All	+0.0401*	+0.0213*	+0.0217	+0.0052	-0.0023	+0.0057

is $+0.01265$, $+0.01730$ and $+0.01675$. With all available rows the all-fold gains are $+0.01388$, $+0.01990$ and $+0.02034$. With all rows the gain rises with every added context; at a matched row count it rises from one context to two and levels off at three. With no context feature, the ridge fit of Equation (3) makes the $3M$ arm equivalent to fitting context-averaged targets at triple weight (the basis and penalty are recomputed for each arm). Against the average single context rather than the best, the spread wins in three of the four folds; the best single context of each fold (Figure 7) was identified after the fact.

The natural-atlas and row-matched unit-direction experiments of Figure 4a answer the same question on response-unseen candidates. Natural atlases add perturbation rows whenever a context is added, whereas the row-matched arm keeps 840 response-unseen training rows on average. Their pooled gains are $+0.0247$, $+0.0421$, and $+0.0492$ across one, two, and three contexts for natural atlases, versus $+0.0123$, $+0.0070$, and $+0.0012$ for row-matched atlases. The same-identity control of Figure 7 fixes the identity universe instead; it uses raw targets and does not mask, so each evaluated candidate’s own training-context rows are in both of its arms (one row in the spread arm, three in the $3M$ arm). Both designs agree that added measurements raise the gain. The context-count comparison at a fixed number of rows depends on the design: under masking with unit-direction targets it falls with each added context, whereas in the unmasked raw-target arms above it rises from one context to two and levels off.

Descriptive least-squares fits over the raw-target natural and row-matched arms of Figure 6, with one intercept per held context, model the gain as $\alpha + \beta_N \log N_{\text{rows}} + \beta_C N_{\text{contexts}}$. They give $\beta_N = +0.00990 \pm 0.00141$ and $\beta_C = +0.00270 \pm 0.00125$. Points within one fold share a held context and a pool, so these standard errors are optimistic. The fit enters the context count linearly and cannot see which contexts are in an arm; neither the same-identity control of Figure 7, which fixes the identity universe, nor the response-unseen comparison of Figure 4a supports a positive context-count effect.

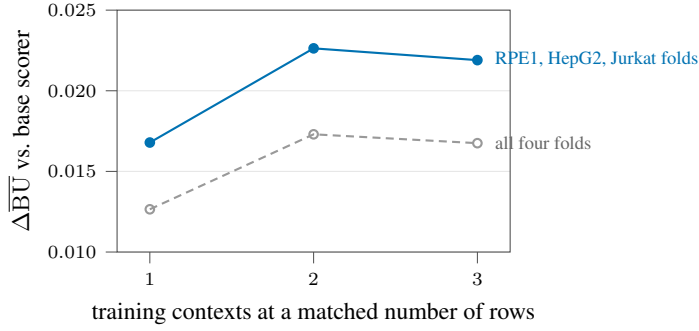


Figure 6: **Row-count matched atlases.** Every arm trains on as many rows as the smallest single-context pool in its fold, split evenly across one, two or three training contexts. Each point is the median paired $\overline{BU}$ difference against the frozen K562 base scorer, over query-view rows, averaged over held-out folds, training-context combinations and five subsampling seeds; point estimates. Blue: the RPE1, HepG2 and Jurkat folds. Unmasked, raw-target counterpart of the dashed arm of Figure 4a, where the gain instead falls. Grey: all four folds, including the K562 fold, whose queries are development-split identities and whose base scorer was trained on K562. The gain rises from one context to two and levels off at three.

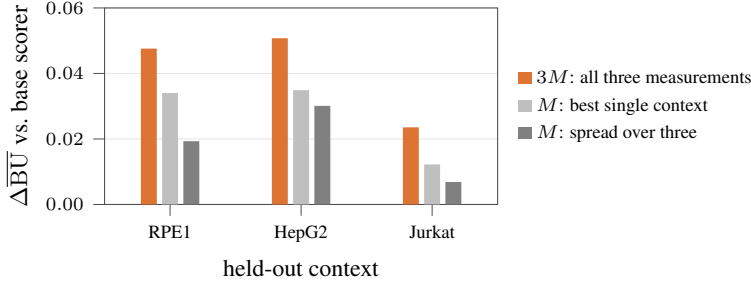


Figure 7: **Same-identity atlas-scale control (raw-target CED).** Median paired $\overline{BU}$ gain of VCDesign-CED over the K562 base scorer, over query-view rows, using only identities measured in all three training contexts. $3M$: all three measurements of each. M : the same number of rows from the best single context of each fold (identified after the fact), or spread over the three contexts (mean of five row-sampling seeds; the other bars are one fit each). Tripling the measurements gains more than spreading them over contexts. No masking is applied, so an evaluated candidate’s own training-context rows are in both arms; the closest response-unseen comparison, which does not fix identities, is Figure 4a. The retrospectively selected best-single arm is a descriptive reference.

D.11 K562 AS A HELD-OUT CONTEXT

K562 behaves differently because the frozen base scorer was trained on K562 responses and is far stronger there: its margin over random is +0.6382, against +0.32 to +0.48 in the other contexts, which leaves the effect branch little room. In every leave-one-context-out, masking, mixed-vocabulary, atlas-scale and vocabulary-size analysis, K562-fold queries are development-split identities and its candidates are K562 training and development identities; the locked evaluation split is used only in Table 1 and Section 5.2. With the three-context atlas the effect branch adds +0.0076 on held K562, and single-context atlases give +0.0071 (Jurkat), −0.0021 (RPE1) and −0.0015 (HepG2) (Table 14). Under controlled masking both arms sit near zero (Table 21): +0.0112 measured and +0.0005 response-unseen, and because these two arms use predictors with different penalties (Section D.8), K562 is compared through the value of historical response exposure, +0.0097 [+0.0075, +0.0123], of the same order as in RPE1 and Jurkat. In mixed vocabularies the gain moves from +0.0112 (all measured) through +0.0073, +0.0051 and +0.0024 to +0.0005 (all response-unseen). In the same-identity atlas-scale control the $3M$ atlas gives +0.00445, the spread atlas −0.00006 and the best single context (Jurkat) +0.00078.

Table 21: **K562 held out under controlled masking (development-split queries)**. Columns as in Table 6. [†]Base scorer trained on K562. [‡]Measured and response-unseen predictors differ in penalty (Section D.8).

Held context (pool)	Measured	Response- unseen	Retained	Value of response exposure [95% CI]
K562 [†] (1,430)	+0.0112	+0.0005	4% [‡]	+0.0097 [+0.0075, +0.0123]

D.12 CONTEXT-SPECIFIC RESPONSE STRUCTURE

On the 468 candidates usable in all four contexts, 0.5655 of the unit-normalized response is context specific. This residual is reproducible across batch-disjoint replicate views (median cosine +0.2419, against −0.0279 for a permuted control), and the median cross-context cosine of one knockdown’s response ranges from +0.1828 (HepG2 and Jurkat) to +0.2658 (RPE1 and HepG2). A cross-fitted, context-specific measured-response oracle exceeds a context-blind one by +0.0948 (HepG2), +0.0807 (Jurkat), +0.0788 (RPE1) and +0.0485 (K562) on this pool. This is the open direction named in Section 6; the context-conditioned effect map cannot learn it from three training contexts (Section E).

E EFFECT-MODEL, SCORER, AND UTILITY VARIANTS

This appendix defines each variant behind Table 7 and Section 5.4, and it reports the variants that the main text only summarizes. Table 23 collects the numbers not shown in Table 7. Unless stated otherwise, a variant changes exactly one component of the method. The frozen K562 base scorer, candidate knowledge, pools, queries, fusion rule, utility and metric stay fixed. Effect-model variants are evaluated in the pooled held-context setting of Section 5.3, where RPE1, HepG2 and Jurkat are held out in turn. There each value is a median paired difference over 1,333 queries (2,666 query-view rows) from 976 query genes, with a 95% interval over query-gene clusters. Scorer, training and utility variants were compared on the development split during method design. The locked K562 result uses raw-target CED; the held-context closure and Figure 3 use the stronger unit-direction refinement.

E.1 EFFECT TARGET AND SCORE FORM

Unit-direction targets. Unit-direction targets give a small refinement that is positive in every held context. The ridge of Equation (3) is refit with each training target replaced by its direction $\mathbf{r}_c/\|\mathbf{r}_c\|$. The response basis $\mathbf{W}$ is rebuilt from the unit targets, and the penalty is re-selected by the same five-fold identity cross-validation. The effect score is the cosine of Equation (4). Fused with the base scorer, unit targets improve on raw targets by +0.0048 [+0.0039, +0.0056] for measured and +0.0042 [+0.0032, +0.0051] for response-unseen candidates. Per context (measured, then response-unseen), the improvement is +0.0045 and +0.0022 in RPE1, +0.0034 and +0.0051 in HepG2, and +0.0053 and +0.0049 in Jurkat, so it is positive in all six cells. Against the base scorer, unit-direction CED adds +0.0470 BU for pooled response-unseen candidates. The refinement holds on the masking pools of 764 to 1,450 candidates. It does not carry over to the 6,255-candidate K562 response-unseen vocabulary of Figure 5. There raw-target CED already gives −0.0103 [−0.0154, −0.0048], and unit-direction CED gives −0.0250.

Keep predicted norm. The predicted norm acts as a context-dependent candidate prior, so VCDDesign-CED uses directional cosine for its effect branch. With unit-direction targets, the score $\|\hat{\mathbf{e}}_c\| \cos(\mathbf{d}_q, \hat{\mathbf{e}}_c)$ keeps the predicted norm as a per-candidate factor and changes nothing else. It helps measured candidates (+0.0142 [+0.0088, +0.0181]) but not response-unseen ones (−0.0004 [−0.0043, +0.0044]). The norm by itself ignores the query. Fused with the base scorer, the unit-target norm alone changes BU by −0.0116 [−0.0180, −0.0062] for response-unseen candidates, whereas the goal-conditioned unit-direction score adds +0.0470. Two scores that carry the norm do gain on pooled rows against raw-target CED. Keeping the norm with unit targets gives +0.0215 [+0.0159, +0.0261] for measured and +0.0069 [+0.0023, +0.0117] for response-unseen candidates. Scoring by the raw-target dot product $\langle \mathbf{d}_q, \hat{\mathbf{e}}_c \rangle$ gives +0.0393 [+0.0322, +0.0451]

and $+0.0295$ [$+0.0240$, $+0.0353$]. A post hoc per-context control shows where these gains come from. Fused with the base scorer, the goal-blind raw predicted norm alone gives $+0.0434$, $+0.0679$ and -0.0790 for response-unseen candidates in RPE1, HepG2 and Jurkat. CED gives $+0.0482$, $+0.0568$ and $+0.0301$ in the same cells (Table 6). The norm is therefore strong in RPE1 and HepG2 and harmful in Jurkat, whereas the goal-conditioned cosine gains in every held context.

Supervision restricted to measurable identities. Breadth of supervision matters more than target cleanliness. This variant restricts the ridge’s training targets to K562 training identities whose two replicate views agree, using the eligibility rule of Section A. The targets are then cleaner but come from fewer identities. On the development split, the effect score alone reaches $\overline{\text{BU}} + 0.0691$, against $+0.1377$ when all training identities supervise the ridge. The typical K562 target is noisy (median cross-view cosine 0.0494 over training identities). Even so, a ridge fit on all of them transfers better than one fit on the reliable subset.

E.2 EFFECT MAP

The linear ridge matches or beats every richer effect map we tested. The first three variants below are compared with the unit-direction ridge, the stronger linear reference, on pooled response-unseen candidates. The context-conditioned map is compared with the raw-target context-blind map in the four leave-one-context-out folds.

Reliability-weighted ridge. Reweighting the ridge by axis reliability has no effect, because reliability is nearly uniform across axes. The variant reweights the ridge penalty per axis of the response basis by the cross-view reliability of that axis, estimated on the training contexts, so that reproducible axes are shrunk less. The change is -0.0001 [-0.0002 , $+0.0000$] $\overline{\text{BU}}$. Inside the rank-256 basis, the median cross-view correlation of an axis is 0.49 to 0.50 for every held context. The top 10% of axes hold 12% of the total reliability, close to the 10% that uniform reliability would give. With reliability this flat, isotropic ridge shrinkage is already close to the reliability-aware solution.

Nonlinear effect memory. A nonlinear effect memory scores below the ridge and matches it only when anchored to it. The model replaces the linear map by a nonlinear predictor. It routes each candidate’s knowledge features to a sparse combination of learned response programs. It also adds responses transported from similar training candidates, which it retrieves from a memory. A candidate never retrieves itself, and response-unseen candidates are removed from the memory as well as from the training rows. Trained end to end, the model changes $\overline{\text{BU}}$ by -0.0253 [-0.0282 , -0.0228] (-0.0233 [-0.0257 , -0.0208] $\text{BU}@20$). The anchored version predicts a sparse correction that is added to the unit-direction ridge, so it starts exactly at the ridge. It matches the ridge, with $+0.0002$ [-0.0007 , $+0.0012$] $\overline{\text{BU}}$ ($+0.0000$ [-0.0010 , $+0.0011$] $\text{BU}@20$). Anchored versions that keep only the parametric program branch or only the memory branch give -0.0016 and -0.0053 $\text{BU}@20$.

Context-invariant adapter. A context-invariant adapter leaves design unchanged, because little of the ridge’s residual is shared across contexts. The adapter corrects the unit-direction ridge’s predicted direction within the tangent space of the unit sphere. Its correction is scaled by a learned scalar initialized at zero, so before training its predictions equal those of the ridge exactly. It is trained either by pooled empirical risk over the training contexts or with an added invariance penalty. The penalty acts on the worst-context excess risk and on the variance of excess risk across contexts. The epoch is selected by leaving one training context out. The adapter gives $+0.0008$ [-0.0003 , $+0.0019$] $\text{BU}@20$ with pooled risk and -0.0012 [-0.0019 , -0.0003] with the invariance penalty. Only 0.315, 0.322 and 0.287 of the ridge’s reproducible residual is shared across the training contexts of RPE1, HepG2 and Jurkat. A correction that must hold across contexts therefore has little to learn.

Context-conditioned effect map. Three training contexts do not identify how responses depend on context. The context-specific structure itself is large and reproducible (Section D.12). This variant adds a low-rank modulation of the effect map by a context representation built from control cells. It is trained on three contexts and evaluated on the fourth. In all four leave-one-context-out folds, the cross-validated penalty shrinks the modulation away, and the difference from the context-blind map is zero to four decimals. A positive control explains this null. We planted a context

residual of the model’s own form, at a share of 0.25, into synthetic responses, with the real Jurkat control cells as the held context. The model recovers it only partly (+0.3885), and the best design value anywhere on the penalty grid is +0.00001.

E.3 RESPONSE BASIS, MODALITY, RANK AND ATLAS SIZE

The ridge depends mainly on STRING and on atlas size, and little on the response basis or the rank. Each variant in Table 22 changes one input of the unit-direction ridge and is scored on pooled response-unseen candidates, fused with the base scorer. No alternative response basis improves on the randomized-SVD basis of Section 4.1 by a margin that matters. The reason is that, out of fold, only 0.012 to 0.018 of the centred candidate-specific response signal is predictable from the knowledge features in any basis. STRING carries most of the transferable signal. Dropping MAP-KG changes BU@20 by -0.0013 , and dropping STRING changes it by -0.0211 . The rank matters little: ranks 64 to 512 move BU@20 by at most 0.0023 from rank 256. Atlas size, rather than context count, tracks the gain. Natural atlases gain more with each added training context. At a matched row count and with these targets, however, added contexts lower the gain. Transfer is positive in 20 of the 21 combinations of held context and source atlas (unit-direction targets, BU@20). The source atlases are every non-empty subset of the held context’s three training contexts, and the one null is HepG2 alone predicting Jurkat. Response-unseen candidates retain 0.76, 1.04 and 0.90 of the measured-candidate gain in RPE1, HepG2 and Jurkat.

Table 22: **Sensitivity of the unit-direction ridge.** Median paired BU@20 differences on pooled response-unseen candidates (RPE1, HepG2 and Jurkat held out in turn), fused with the base scorer, with 95% intervals over query-gene clusters where recorded. Atlas arms average per query over the atlases in each class; row-matched arms subsample every atlas to the smallest single-context atlas. The transfer row counts, per held context, the source atlases formed by every non-empty subset of its three training contexts (21 cells, each against the base scorer). Retention is the response-unseen over the measured BU@20 gain against the base scorer. STRING and atlas size move the gain; the response basis and the rank barely do.

Variant	Reference	Result
Shared-covariance basis	Unit-direction ridge	-0.0050 [$-0.0072, -0.0034$]
Predictive basis	Unit-direction ridge	$+0.0009$ [$+0.0006, +0.0013$]
Shared and predictive basis combined	Unit-direction ridge	$+0.0013$ [$-0.0003, +0.0030$]
Drop MAP-KG (STRING only)	Unit-direction ridge	-0.0013 [$-0.0026, -0.0002$]
Drop STRING (MAP-KG only)	Unit-direction ridge	-0.0211 [$-0.0235, -0.0177$]
Rank 64 / 128 / 512	Rank 256	-0.0023 / -0.0006 / $+0.0008$
Natural atlas, 1 training context	Base scorer	$+0.0247$ [$+0.0212, +0.0273$]
Natural atlas, 2 training contexts	Base scorer	$+0.0421$ [$+0.0378, +0.0457$]
Natural atlas, 3 training contexts	Base scorer	$+0.0492$ [$+0.0456, +0.0534$]
Row-matched, 1 training context	Base scorer	$+0.0123$ [$+0.0089, +0.0154$]
Row-matched, 2 training contexts	Base scorer	$+0.0070$ [$+0.0046, +0.0095$]
Row-matched, 3 training contexts	Base scorer	$+0.0012$ [$-0.0011, +0.0036$]
Transfer, held context $\times$ source subset	Base scorer	positive in 20 of 21
Retention, RPE1 / HepG2 / Jurkat	Measured candidates	0.76 / 1.04 / 0.90
Win rate, RPE1 / HepG2 / Jurkat	Base scorer	0.805 / 0.877 / 0.740

Table 23: **Effect-model, scorer and utility variants not shown in Table 7.** Each row gives the variant and, in parentheses, its reference. Held-context rows are median paired differences with 95% intervals over query-gene clusters, pooled over 1,333 queries (2,666 query-view rows) unless a context is named; rows without an interval are point estimates. Development-split rows are median paired differences, except rows written “a vs b”, which give the levels of the variant and its reference. Pairs a / b give two regimes or two grading directions as labelled in the text. Measured resp.: rankings built from measured responses, nothing trained. Real. obj.: graded by the four-quarter realized objective. For the decision-aligned sampler the interval is on the mean over query identities. Richer maps, a larger scorer and other utilities do not improve the ranking.

Variant (reference)	Setting	Metric	Result
<i>Score composition and effect target</i>			
Effect score alone (base scorer)	Pooled, measured	$\overline{BU}$	-0.0272 [-0.0439, -0.0116]
Effect score alone (base scorer)	Pooled, unseen	$\overline{BU}$	-0.1003 [-0.1137, -0.0831]
Goal-blind norm, unit targets (base)	Pooled, unseen	$\overline{BU}$	-0.0116 [-0.0180, -0.0062]
Unit-direction CED (base scorer)	Pooled, unseen	$\overline{BU}$	+0.0470
Unit targets, effect only (raw targets)	Pooled, measured	$\overline{BU}$	+0.0396 [+0.0362, +0.0433]
Unit targets, effect only (raw targets)	Pooled, unseen	$\overline{BU}$	+0.0262 [+0.0236, +0.0305]
Keep predicted norm (unit targets)	Pooled, measured	$\overline{BU}$	+0.0142 [+0.0088, +0.0181]
Keep predicted norm (unit targets)	Pooled, unseen	$\overline{BU}$	-0.0004 [-0.0043, +0.0044]
Keep norm, unit targets (raw targets)	Pooled, measured	$\overline{BU}$	+0.0215 [+0.0159, +0.0261]
Keep norm, unit targets (raw targets)	Pooled, unseen	$\overline{BU}$	+0.0069 [+0.0023, +0.0117]
Raw-target dot product (raw targets)	Pooled, measured	$\overline{BU}$	+0.0393 [+0.0322, +0.0451]
Raw-target dot product (raw targets)	Pooled, unseen	$\overline{BU}$	+0.0295 [+0.0240, +0.0353]
Raw norm alone, post hoc (base)	RPE1, unseen	$\overline{BU}$	+0.0434
Raw norm alone, post hoc (base)	HepG2, unseen	$\overline{BU}$	+0.0679
Raw norm alone, post hoc (base)	Jurkat, unseen	$\overline{BU}$	-0.0790
Unit-direction targets (raw targets)	RPE1	$\overline{BU}$	+0.0045 / +0.0022
Unit-direction targets (raw targets)	HepG2	$\overline{BU}$	+0.0034 / +0.0051
Unit-direction targets (raw targets)	Jurkat	$\overline{BU}$	+0.0053 / +0.0049
Unit-direction CED (base scorer)	K562, 6,255 unseen	$\overline{BU}$	-0.0250
Raw-target CED (base scorer)	K562, 6,255 unseen	$\overline{BU}$	-0.0103 [-0.0154, -0.0048]
<i>Effect map (unit-direction ridge; pooled, response-unseen)</i>			
Reliability-weighted ridge	Pooled	$\overline{BU}$	-0.0001 [-0.0002, +0.0000]
Nonlinear effect memory	Pooled	$\overline{BU}$	-0.0253 [-0.0282, -0.0228]
Nonlinear effect memory	Pooled	$BU@20$	-0.0233 [-0.0257, -0.0208]
Nonlinear memory, anchored to ridge	Pooled	$\overline{BU}$	+0.0002 [-0.0007, +0.0012]
Nonlinear memory, anchored to ridge	Pooled	$BU@20$	+0.0000 [-0.0010, +0.0011]
Memory, parametric branch, anchored	Pooled	$BU@20$	-0.0016
Memory, memory branch, anchored	Pooled	$BU@20$	-0.0053
Invariant adapter, pooled risk	Pooled	$BU@20$	+0.0008 [-0.0003, +0.0019]
Invariant adapter, invariance penalty	Pooled	$BU@20$	-0.0012 [-0.0019, -0.0003]
<i>Context-conditioned map (raw-target context-blind map; four leave-one-context-out folds)</i>			
Context-conditioned map (context-blind)	Four held contexts	$\overline{BU}$	0.0000 in every fold
Planted residual, share 0.25	Jurkat held	recovery	+0.3885
Planted residual: best design value	Jurkat held	$\overline{BU}$	+0.00001
<i>Development split</i>			
Measurable-identity targets (all)	Effect score alone	$\overline{BU}$	+0.0691 vs +0.1377
Decision-aligned sampler (standard)	Text-aug. ranker	$\overline{BU}$	-0.0794; mean -0.0820 [-0.0920, -0.0720]
Three-way scorer (bilinear scorer)	Capacity test	pair acc.	0.5261 vs 0.5252
Factorial probe, measured responses	Probe scorer	pair acc.	0.5871 / 0.5883
Factorial probe, knowledge features	Probe scorer	pair acc.	0.5228 / 0.5245
Latent endpoint utility (cosine)	Measured resp.	$\overline{BU}$	-0.1222
Attenuation-aware ranking (direction)	Measured resp.	real. obj. $\overline{BU}$	-0.1778 / -0.2662
Kernel distributional transition	Measured resp.	real. obj. $\overline{BU}$	-0.0202 / -0.0133
Distributional tie-breaking (mean only)	Measured resp.	real. obj. $\overline{BU}$	+0.000018

E.4 SCORER, TRAINING, AND UTILITY

Decision-aligned pair sampling. Sampling the pairs that decide the top of the ranking lowered design value. The pairwise loss of the base scorer samples candidate pairs from the utility top, the predicted top and random candidates. This sampler covers only a small share of the pairs that decide the top of the ranking (median decision-mass coverage 0.0260). A decision-aligned sampler targets those pairs instead. It was continued from the text-augmented ranker checkpoint under an otherwise identical schedule. It changes development-split $\overline{\text{BU}}$ by a median of -0.0794 (mean -0.0820 $[-0.0920, -0.0720]$, bootstrap over query identities). In this run, targeting the hardest pairs cost more elsewhere than it gained.

Three-way scorer. The trilinear scorer we tested adds capacity but leaves pair accuracy unchanged, whereas candidate information moves it. This scorer models an explicit source-by-target-by-candidate interaction, with twice the parameters of the bilinear form and five times the training budget. On the development split it reaches a uniform pair accuracy of 0.5261, against 0.5252 for the bilinear form. Within the factorial probe of Section D.3, uniform pair accuracy is 0.5228 and 0.5245 (learned query encoding and raw transition) when the probe reads knowledge features, including text. It is 0.5871 and 0.5883 when the probe reads measured responses instead.

Utility and score form. Four alternatives to the direction-cosine utility and the pseudobulk transition score lower or change nothing under the available measurements. We therefore use the signed direction cosine of pseudobulk transitions as the benchmark utility and effect score. Development-split rankings use measured responses; no model is trained. A latent-endpoint utility changes $\overline{\text{BU}}$ by a median of -0.1222 against the cosine, and 80.8% of its variance does not depend on the query. The other three alternatives are graded by the four-quarter realized objective (Section A.1). An attenuation-aware objective that also selects per-candidate knockdown strength changes realized-objective $\overline{\text{BU}}$ by medians of -0.1778 and -0.2662 in the two grading directions. Kernel distributional transitions change it by -0.0202 and -0.0133 ; using distributional scores only to break mean-ordering ties changes it by $+0.000018$ (mean -0.000118) in the first direction. These comparisons diagnose measurement resolution; they do not establish directional cosine as a universal biological objective.